

UNIT – 1: BIOMOLECULES-PART-1

Points to be covered in this topic

- ❖ Introduction of Biochemistry
- ❖ Carbohydrate
 - ➤ Introduction
 - ➤ Classification
 - ➤ Chemical nature
 - ➤ Biological role
- ❖ Lipids
 - ➤ Introduction
 - ➤ Classification
 - ➤ Chemical nature
 - ➤ Biological role

BIOMOLECULES

❖ INTRODUCTION

- Biochemistry is a discipline of Chemistry that deals with the **chemical composition of living organisms**.

- Biochemistry can be simply defined as, "chemistry of the living cell "
- The term **Biochemistry** was introduced by **Carl Neuberg in 1903**.
- The **living matter** is composed of mainly **six elements** — carbon, hydrogen, oxygen, nitrogen, **phosphorus** and **sulfur**.
- These elements together constitute about **90% of the dry weight** of the human body.
- Several other functionally important elements are also found in the cells.

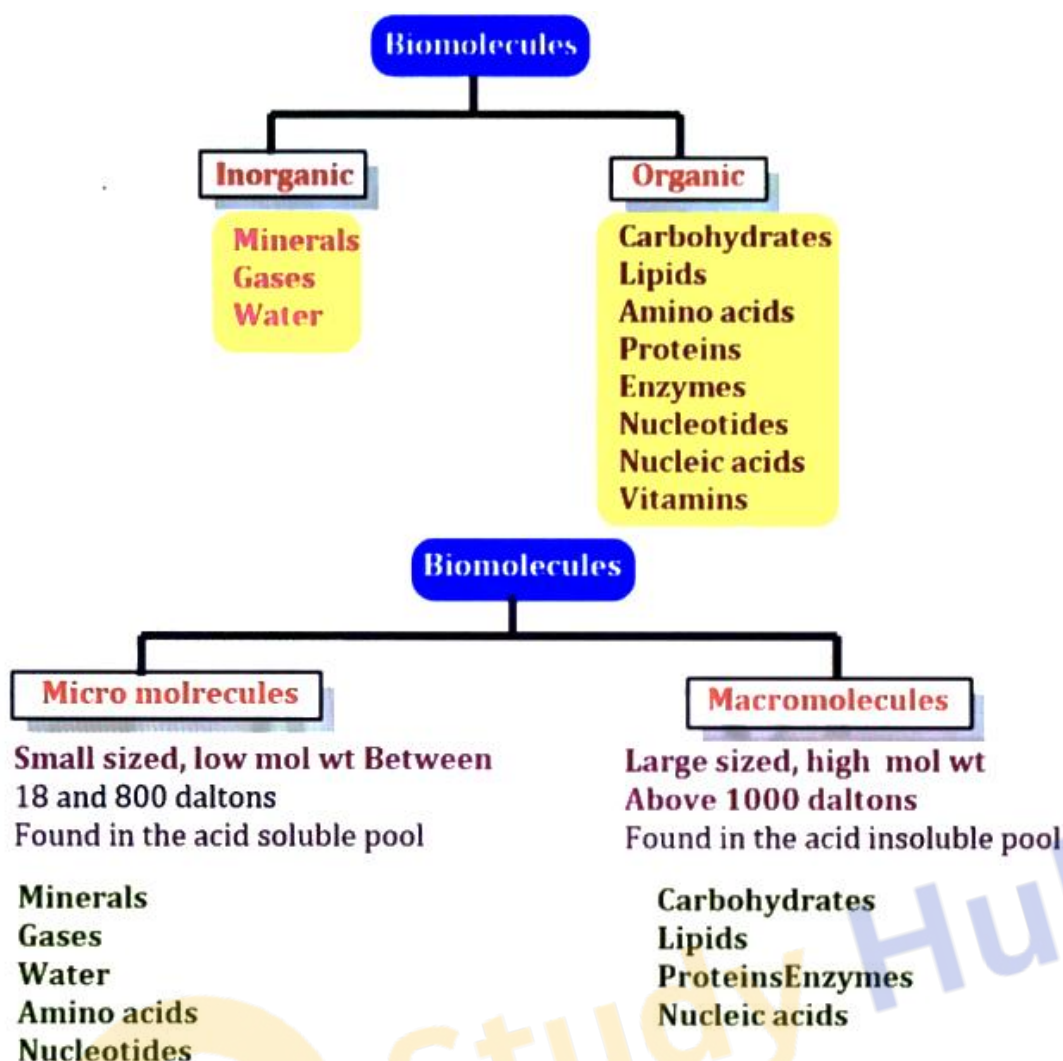


Definition

- **Biochemistry** is the study of **structures and the interactions of biological macromolecules**.
- These macromolecules include **protein, nucleic acids, lipids, and carbohydrates** present in body.

Biomolecules

- Biomolecules are the **most essential organic molecules**, which are involved in the **maintenance and metabolic processes of living organisms**.
- These non-living molecules are the actual foot-soldiers of the battle of sustenance of life.
- They range from **small molecules** such as **primary and secondary metabolites and hormones** to **large macromolecules** like proteins, nucleic acids, carbohydrates, lipids etc.



The major complex biomolecules of cells

S.NO.	BIOMOLECULE	BUILDING BLOCK (REPEATING UNIT)	MAJOR FUNCTIONS
1.	Protein	Amino acids	Fundamental basis of structure and function of cell (static and dynamic functions).
2.	Deoxyribonucleic acid (DNA)	Deoxyribonucleotides	Repository of hereditary information.

S.NO.	BIOMOLECULE	BUILDING BLOCK (REPEATING UNIT)	MAJOR FUNCTIONS
3.	Ribonucleic acid (RNA)	Ribonucleotides	Essentially required for protein biosynthesis .
4.	Polysaccharide (glycogen)	Monosaccharides (glucose)	Storage form of energy to meet short term demands.
5.	Lipid	Fatty acids, glycerol	Storage form of energy to meet long term demands; structural components of membranes.

CARBOHYDRATE

❖ INTRODUCTION TO CARBOHYDRATES

- Carbohydrates are **organic compounds** with general formula $C_n(H_2O)_n$.
- They are composed of **carbon, hydrogen, and oxygen** having the ratio of **oxygen and hydrogen atom 2:1**. Eg.: Glucose ($C_6H_{12}O_6$)
- Carbohydrates are the **primary source of energy** as cells utilize carbohydrate directly for **cellular respiration in the presence of oxygen**.
- At the end of reaction, carbon dioxide, water, and energy is obtained (in the form of ATP), as shown below:

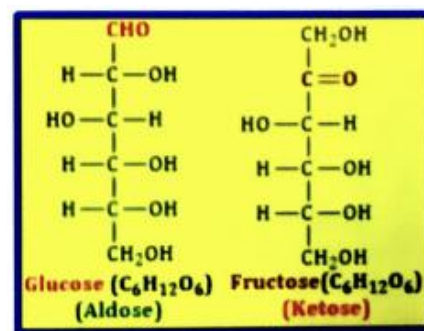


Definition to carbohydrates

- Carbohydrates may be defined chemically as **aldehyde or ketone derivatives of polyhydroxy (more than one hydroxy group) alcohols** or as compounds that yield these derivatives on **hydrolysis**.

✓ Carbohydrates contain the following functional groups:

- (1) Alcoholic hydroxyl groups (-OH),
- (2) Aldehyde group (CHO), and
- (3) Ketone group (C=O)



Functions of carbohydrates

✓ Carbohydrates have a wide range of functions. The following are few of them:

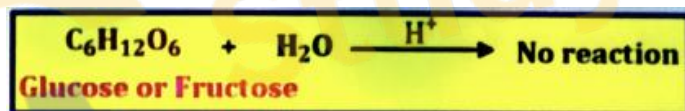
- **Source of energy** for living beings, e.g. glucose.
- **Storage form of energy**, e.g. glycogen in animal tissue and starch in plants.
- **Serve as structural component**, e.g. glycosaminoglycans in humans, cellulose in plants and chitin in insects.
- **Non-digestible carbohydrates** like cellulose, serve as dietary fibers.
- Constituent of nucleic acids RNA and DNA, e.g. ribose and deoxyribose sugar.
- Play a role in lubrication, **cellular intercommunication and immunity**.
- Carbohydrates are also involved in **detoxification**, e.g. glucuronic acid.

❖ CLASSIFICATION OF CARBOHYDRATES

CLASSIFICATION	TYPES	EXAMPLE
MONOSACCHARIDES	Triose ($C_3H_6O_3$)	Glyceraldehyde, Dihydroxyacetone
	Tetrose ($C_4H_8O_4$)	Erythrose, Erythrulose
	Pentose ($C_5H_{10}O_5$)	Ribose, Ribulose, Deoxyribose, Xylose, Xylulose
	Hexose ($C_6H_{12}O_6$)	Glucose, Galactose, Mannose, Fructose
	Heptose ($C_7H_{14}O_7$)	D-Sedoheptulose
DISACCHARIDES		Sucrose, Lactose, Maltose
TRISACCHARIDES		Raffinose, Gentianose
POLYSACCHARIDES	Homopolysaccharide	Starch, Dextrin, Inulin, Glycogen, Cellulose
	Heteropolysaccharides	Hyaluronic acid, Chondroitin, Heparin, Keratan sulphate, Dermatan sulphate.

Monosaccharides (greek: mono = one)

- These are unit carbohydrates with **polyhydroxy aldehydes or ketones**.
- These sugars are **simplest and non-hydrolysable**.
- E.g. Glucose and Fructose.



✓ PROPERTIES

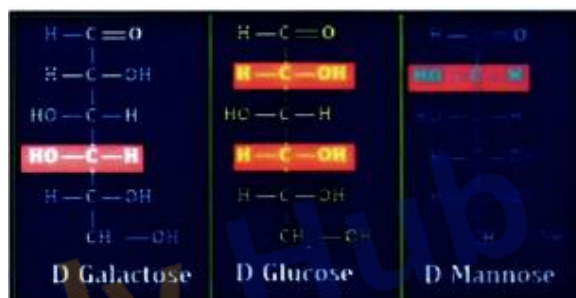
- They are a **sweet taste**, crystalline in character and **soluble in water**.
- **Action of Strong Acids**: Furfural formation.
- **Action of Alkalies**: Enolization.
- **Oxidation**: Sugar acid formation.
- **Reduction**: Sugar alcohol formation.
- **Action of phenylhydrazine**: Osazone formation.



✓ STEREOISOMER OF CARBOHYDRATE

1. Epimer

- When two monosaccharides **differ from each other in configuration around single asymmetric carbon** (other than anomeric carbon) atom they are referred to as **epimer of each other**.
- Example:** Glucose and Galactose are **C₄ epimer** while Glucose and Mannose are **C₂ epimer**.



2. Anomers

- The **α and β form of D-glucose** are known as **Anomer**. They differ from each other in the configuration only around **C₁** known as **anomeric carbon**.
- Example :** **α - D Glucose** and **β - D glucose**, **α - D Fructose** and **β -D fructose**

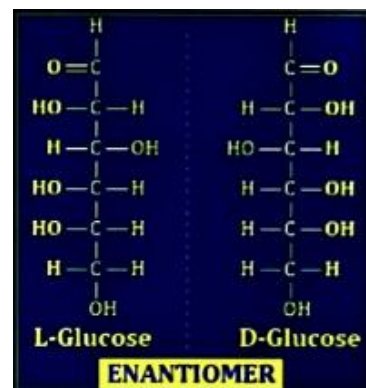
3. Mutarotation



- It is defined as the change in the **specific optical rotation** by the interconversion of **α & β form of D-glucose** to an equilibrium mixture.
- Mutarotation is studied by the **measuring the rotation of plane polarized light**.

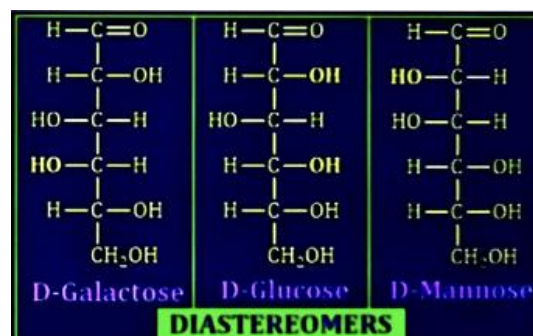
4. Enantiomer

- Enantiomer are special type of stereoisomer that are **mirror image of each other**.
- Example:** D-Glucose and L-Glucose and D-Fructose and L-Fructose.



5. Diastereomers

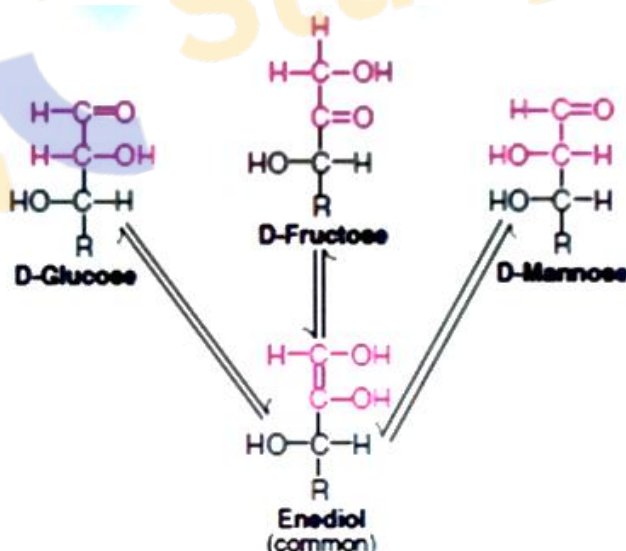
- The diastereomer is used to represent the stereoisomer that are **not mirror image of each other**.
- Example:** D-Mannose and D-Galactose.



✓ Reactions of monosaccharides

Tautomerization or enolization

- The process of shifting a hydrogen atom from one carbon atom to another to produce enediols is known as **tautomerization**. Sugars possessing anomeric carbon atom undergo tautomerization in alkaline solutions.



Formation of a common enediol from glucose, fructose and mannose (R corresponds to the end 3 carbon common structure).

Reducing properties

- The sugars are classified as **reducing or nonreducing**. The reducing property is attributed to the **free aldehyde or keto group of anomeric carbon**.

Reducing and non-reducing sugar

REDUCING SUGAR	NON REDUCING SUGAR
Carbohydrate with free aldehyde or free ketone	Free functional groups are not available
They are hemiacetal or hemiketal form	They are in acetal or ketal form
Exhibit mutarotation	Not exhibit mutarotation
Do form osazones with phenyl hydrazine	Do not form osazones
Reducing sugar reduce cupric ion of copper sulphate to cuprous ion yellow precipitate of Benedict, Fehling & Barfoed's test	Do not reduce cupric ion of Benedict solution

Example (Reducing): Glucose, Fructose, Maltose, Lactose.

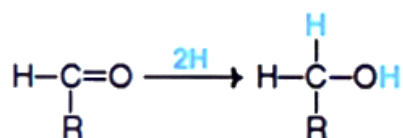
Example (Non Reducing): Sucrose, Trehalose, Starch.

Oxidation

- Depending on the oxidizing agent used, the **terminal aldehyde (or keto) or the terminal alcohol or both the groups may be oxidized**. For instance, consider glucose:
 - Oxidation of aldehyde group ($\text{CHO} \rightarrow \text{COOH}$)** results in the formation of gluconic acid.
 - Oxidation of terminal alcohol group ($\text{CH}_2\text{OH} \rightarrow \text{COOH}$)** leads to the production of glucuronic acid.

Reduction

- When treated with reducing agents such as **sodium amalgam**, the aldehyde or **keto group of monosaccharide is reduced to corresponding alcohol**, as indicated by the general formula:



- ✓ The important monosaccharides and their corresponding alcohols are given below.

- D-Glucose $\rightarrow$ D-Sorbitol
- D-Galactose $\rightarrow$ D-Dulcitol
- D-Mannose $\rightarrow$ D-Mannitol
- D-Fructose $\rightarrow$ D-Mannitol + D-Sorbitol
- D-Ribose $\rightarrow$ D-Ribitol

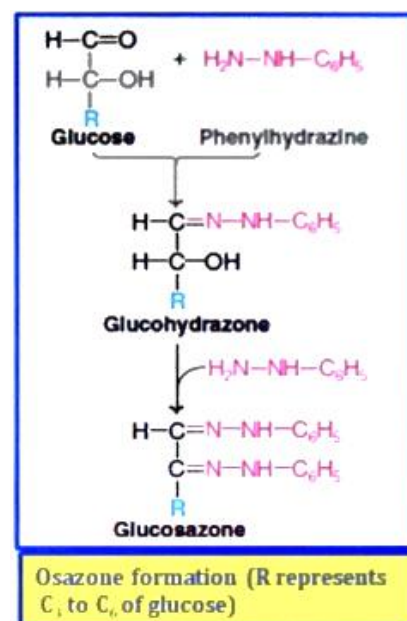
Dehydration

- When treated with concentrated sulfuric acid, **monosaccharides undergo dehydration with an elimination of 3 water molecules**. Thus hexoses give **hydroxymethyl furfural** while pentoses give **furfural** on dehydration.
- These furfurals can condense with phenolic compounds (α -naphthol) to form coloured products. This is the chemical basis of the popular Molisch test.
- Bial's test**: Pentoses react with **strong HCl** to form **furfural derivatives which in turn react with orcinol to form green coloured complex**. Bial's test is useful for detection of xylose in urine in essential pentosuria.

- **Mucic acid test:** Galactose when treated with nitric acid forms insoluble mucic acid crystals.

Osazone formation

- **Phenylhydrazine in acetic acid**, when boiled with reducing sugars, forms osazones in a reaction.
- As is evident from the reaction, the first two carbons (**C₁** and **C₂**) are involved in osazone formation.



Formation of Esters

- The alcoholic groups of monosaccharides may be esterified by **non-enzymatic** or **enzymatic** reactions.
- **Glucose 6-phosphate** and **glucose 1-phosphate** are good examples. **ATP** donates the phosphate moiety in ester formation.

Oligosaccharides (Greek: oligo-few)

- **Oligosaccharides** contain **2-10 monosaccharide** molecules which are liberated on **hydrolysis**.
- Based on the number of **monosaccharide** units present, the **oligosaccharides** are further subdivided into **disaccharides**.
- **E.g., Sucrose, Lactose, Maltose**

CHEMICAL PROPERTIES

- They have a sweet taste.
- They are crystalline in character.
- They are soluble in water.



Oligosaccharides are classified into

1. **Disaccharides**: These are sugars which hydrolyze into **2 molecules** of the same or different monosaccharides.
 - For example, **sucrose** hydrolyzes into a molecule of each **glucose** (aldose) and **fructose** (ketose).
2. **Trisaccharides**: These sugars are hydrolyzed into **3 molecules** of monosaccharides.
 - For example, **raffinose** ($C_{18}H_{32}O_{16}$) is hydrolyzed into **galactose**, **glucose**, and **fructose**.
3. **Tetrasaccharides**: These sugars are hydrolyzed into **4 molecules** of monosaccharides.

OLIGOSACCHARIDES	
Disaccharides	Sucrose, Lactose, Maltose
Trisaccharides	Raffinose, Gentianose

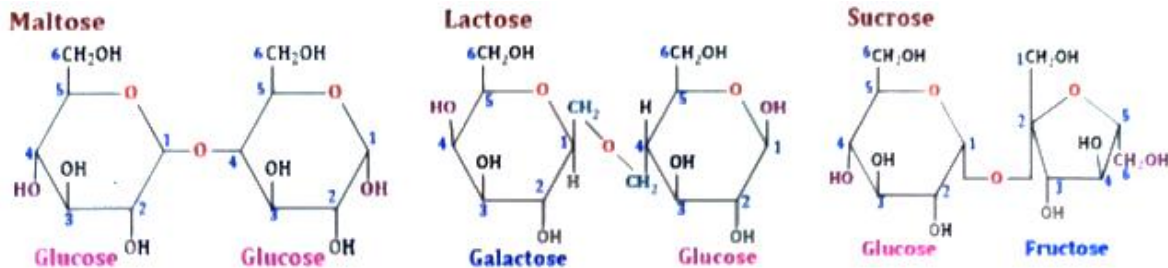
✓ DISACCHARIDES

- Among the **oligosaccharides**, **disaccharides** are the most common.
- As is evident from the name, a **disaccharide** consists of **two monosaccharide units** (**similar** or **dissimilar**) held together by a **glycosidic bond**.
- They are **crystalline**.
- They are **water-soluble**.
- They are **sweet to taste**.

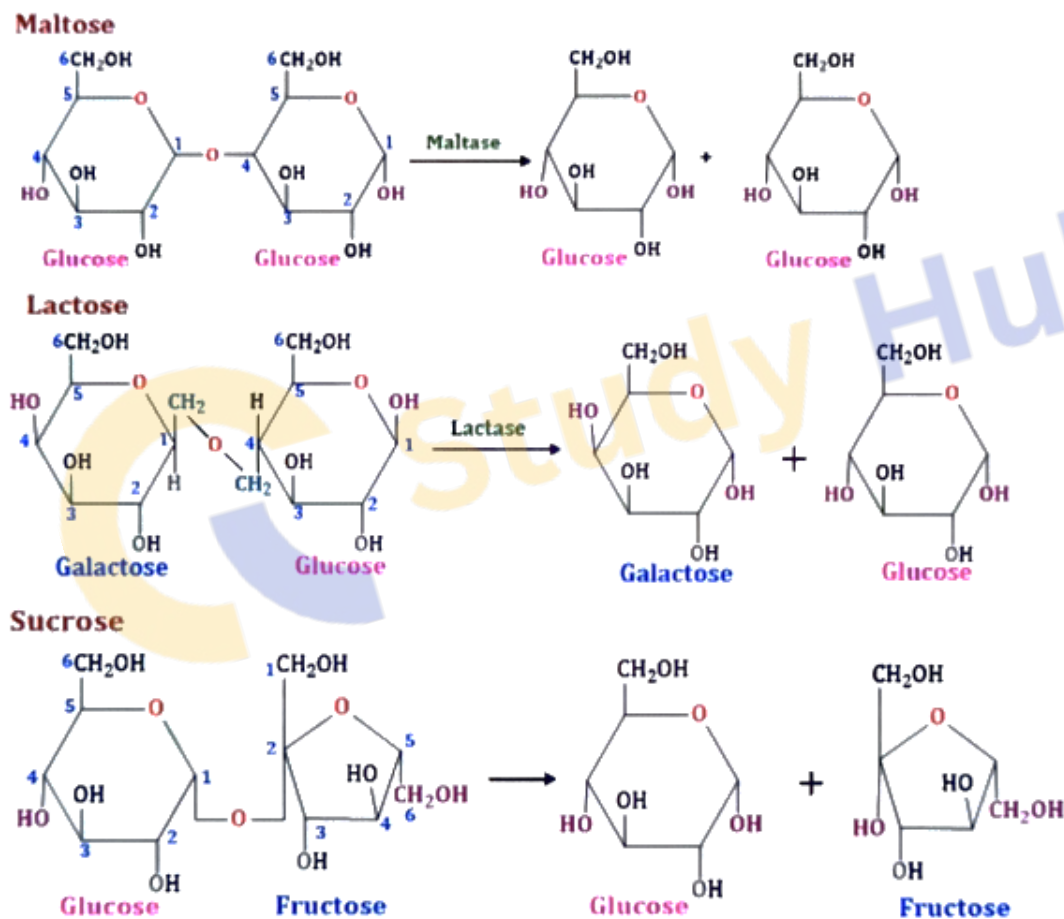
THE DISACCHARIDES ARE OF TWO TYPES

1. **Reducing disaccharides** with free **aldehyde** or **keto** group, e.g., **maltose**, **lactose**.
2. **Non-reducing disaccharides** with no free **aldehyde** or **keto** group, e.g., **sucrose**, **trehalose**.

CARBOHYDRATES	SUGAR UNIT	LINKAGE
Sucrose	α - D Glucose + β -D Fructose	α, β (1 $\rightarrow$ 2)
Lactose	β -D- Glucose + β -D- Galactose	β - (1 $\rightarrow$ 4)
Maltose	α -D- Glucose + α -D- Glucose	α - (1, 4)



✓ Breakdown of disaccharides



Polysaccharides (Greek: poly-many) or Glycans

- **Polysaccharides** are polymers consisting of **hundreds or thousands** of monosaccharide units.

- They are also called **glycans** or **complex carbohydrates**.
- They may be either **linear** (e.g., **cellulose**) or **branched** (e.g., **glycogen**) in structure.
- **Polysaccharides** have **high molecular weight** and are only sparingly soluble in water.
- They are **not sweetish** and do not exhibit any of the properties of **aldehyde or ketone group**.

✓ **Polysaccharides are of two types**

- Homopolysaccharides (Homoglycans)**
- Heteropolysaccharides (Heteroglycans)**

1. Homopolysaccharides (Homoglycans)

- When a polysaccharide is made up of several units of one and the **same type of monosaccharide unit**, it is called **homopolysaccharide**.
- The most common **homoglycans** are: **Starch, Dextrin, Glycogen, Insulin, and Cellulose**.

2. Heteropolysaccharides (Heteroglycans)

- They contain **two or more** different types of **monosaccharide units** or their derivatives.
- **Heteropolysaccharides** present in human beings is **glycosaminoglycans** (mucopolysaccharides), e.g., **Heparin, Chondroitin sulfate, Hyaluronic acid, Dermatan sulfate, Keratan sulfate, Blood group polysaccharides**.

Classification of polysaccharides

CLASSIFICATION	TYPES	EXAMPLE
Polysaccharides	Homopolysaccharide	Starch, Dextrin, Inulin, Glycogen, Cellulose
	Heteropolysaccharides	Hyaluronic acid, Chondroitin, Heparin, Keratan sulfate, Dermatan sulphate

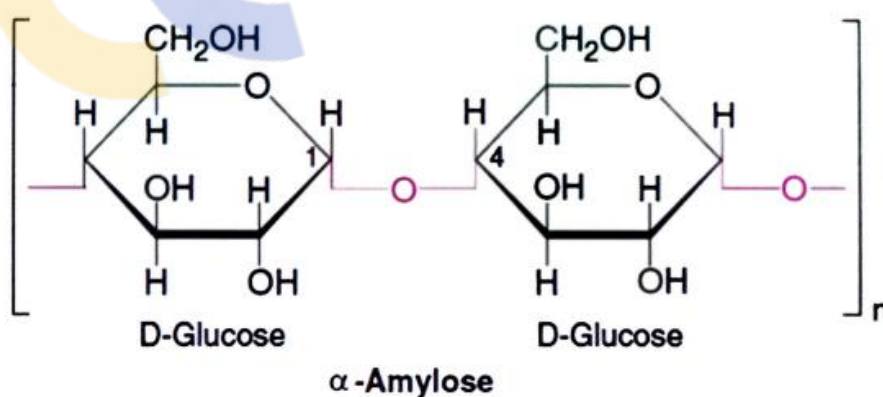
HOMOPOLYSACCHARIDE

✓ STARCH

- It is the storage form of **glucose** in plants, e.g., in **potato**, in **grains** and **seeds** and in many fruits.
- Starch is composed of **two constituents** viz. **amylose** and **amylopectin**.

✓ AMYLOSE

- Amylose** is a linear polymer of **D-glucose** units joined by $\alpha - 1 \rightarrow 4$ glycosidic linkages.
 - Amylose** (15-20%) is **soluble** in water.

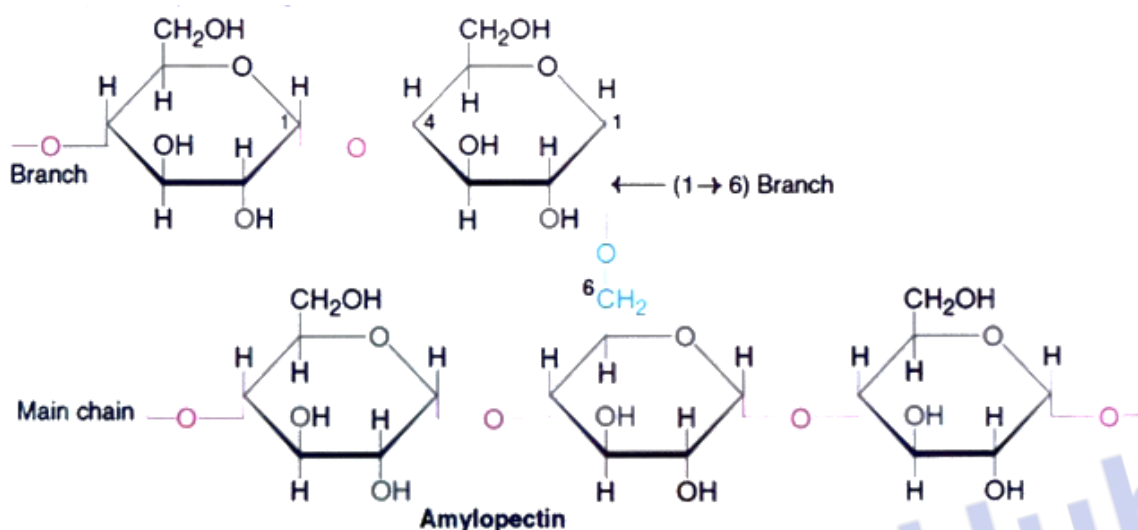


✓ AMYLOPECTIN

- Amylopectin** is **insoluble** in water (**80-85%**).



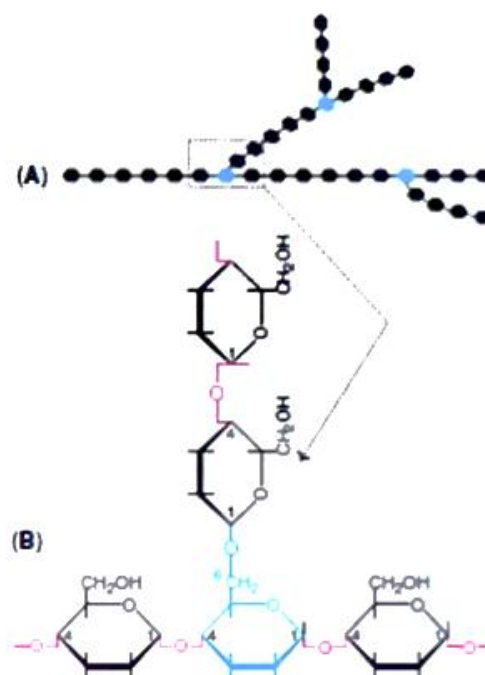
- **Amylopectin** is structurally identical to those of **amylose** ($\alpha - 1 \rightarrow 4$) glycosidic linkages but with side chains joining them by ($\alpha - 1 \rightarrow 6$) linkages.
- Thus, **amylopectin** is a branched polymer having both $\alpha - (1 \rightarrow 4)$ and $\alpha - (1 \rightarrow 6)$ linkages.



- The branch points in amylopectin are created by $\alpha - 1 \rightarrow 6$ bonds and occur at an interval of **20 to 30** units of glucose.

✓ GLYCOGEN (ANIMAL STARCH)

- **Glycogen** is the major storage form of **carbohydrate** (**glucose**) in animals, found mostly in **liver** and **muscle**.
- It is often called **animal starch**.
- The structure of **glycogen** is similar to that of **amylopectin**, except that it is **more highly branched**, having $\alpha - (1 \rightarrow 6)$ linkages at intervals of about **8 to 10 glucose units**.



- **Glucose** is the repeating unit in **glycogen** joined together by $\alpha(1\rightarrow4)$ glycosidic bonds, and $\alpha(1\rightarrow6)$ glycosidic bonds at branching points.

✓ **Function**

- The function of **muscle glycogen** is to act as a readily available source of **glucose** for **energy** within muscle itself.
- **Liver glycogen** is concerned with **storage** and **maintenance** of the blood glucose.

✓ **DEXTRINS**

- **Dextrins** are the breakdown products of **starch** by the enzyme **amylase** or dilute acids.
- The various intermediates (identified by iodine coloration) are soluble starch (**blue**), amylopectin (**violet**), erythrodextrin (**red**) and achrodextrin (no colour).

✓ **DEXTRANS**

- **Dextrans** are **polymers of glucose**, produced by microorganisms. They are used as **plasma volume expanders** in **transfusion**, and **chromatography**. (e.g. gel filtration).

✓ **INULIN**

- Inulin is a **polymer of fructose i.e., fructosan**. It occurs in dahlia bulbs, garlic, onion etc.
- It is used for assessing kidney function through measurement of glomerular filtration rate (GFR).

✓ **CELLULOSE**

- Cellulose occurs **exclusively in plants** and it is the most abundant organic substance in plant kingdom.

- Cellulose is composed of **β -D-glucose units linked by β (1 $\rightarrow$ 4) glycosidic bonds.**

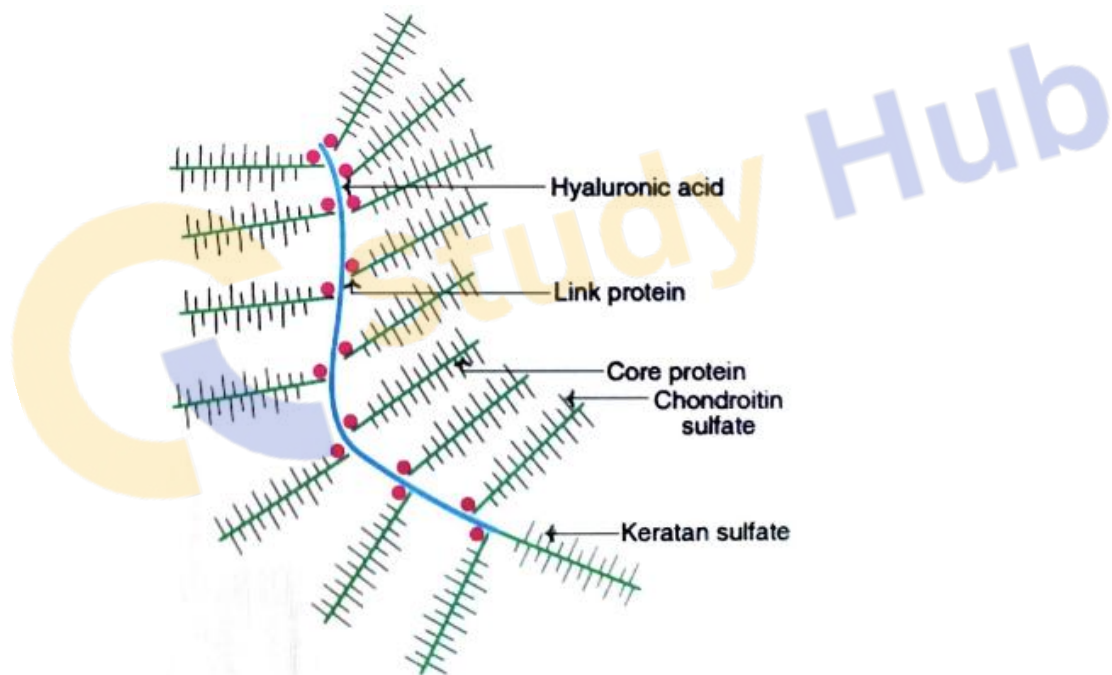
HOMOPOLYSACCHARIDE

✓ MUCOPOLYSACCHARIDES

- Mucopolysaccharides are heteroglycans made up of **repeating units of sugar derivatives, namely amino sugars and uronic acids.**
- These are more commonly known as **glycosaminoglycans (GAG).**

✓ HYALURONIC ACID

- Hyaluronic acid serves as a **lubricant and shock absorbant in joints.**

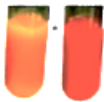


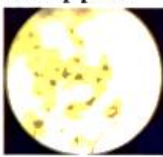
Diagrammatic representation of a proteoglycan complex

CARBOHYDRATES AND THEIR BIOLOGICAL IMPORTANCE

MONOSACCHARIDES	BIOLOGICAL IMPORTANCE
D-Ribose	Structural element of nucleic acid and coenzyme. Example: ATP, NAD, NADP.
D-Ribulose	Intermediate in Hexose Monophosphate Shunt.
D-Xylulose	Intermediate in Uronic acid pathway.
D-Lyxose	A constituent of Lyxoflavin of heart muscle.
D-Glucose	It is carried by the blood and used by the tissue. Its presence in the urine cause glycosuria.
D-Galactose	Synthesized in mammary gland to make the lactose of milk. It is constituent of glycolipid and glycoprotein.
D-Mannose	Constituent of prosthetic polysaccharides of albumin, glycoprotein and blood group.
D-Xylose	Constituent of glycoproteins and gums.

QUALITATIVE TESTS

SNO.	TEST	OBSERVATION	INFERENCE
1	Molisch test α - naphthol in ethanol + conc. H ₂ SO ₄	Purple red ring at junction of two liquid 	For all carbohydrate
2	Benedict's test Copper(II) sulfate pentahydrate + Sodium carbonate + Sod. Citrate + boil for 2 minute.	Orange red or brick red Precipitate 	Reducing sugar
3	Fehling's test Fehling solution A + Fehling solution B and boil for 2 min.	Yellow red precipitate 	Reducing sugar
4	Tommerstest Tommers reagent and boil for 2 min.	Yellow or red precipitate 	Reducing sugar

SNO.	TEST	OBSERVATION	INFERENCE
5	Barfoed's test Neutral copper acetate in acetic acid and boil on water bath	Bricks-Red precipitate at bottom Of the test tube 	Distinguish between monosaccharide form disaccharide
6	Seliwanoff's test Resorcinol in HCl	Deep cherry Red colored complex 	Distinguish between ketose and aldose form
7	Osazone test Phenyl hydrazine + sod. Acetate+ water + heat for 20 min.	Greenish yellow needle/ Broom stick shape crystal 	Test for glucose, Fructose, Mannose
		Puff shaped crystal 	Test for lactose
		Patel shaped crystals appear like sun flower 	Test for maltose

IMPORTANT CARBOHYDRATES AND THEIR IMPORTANCE

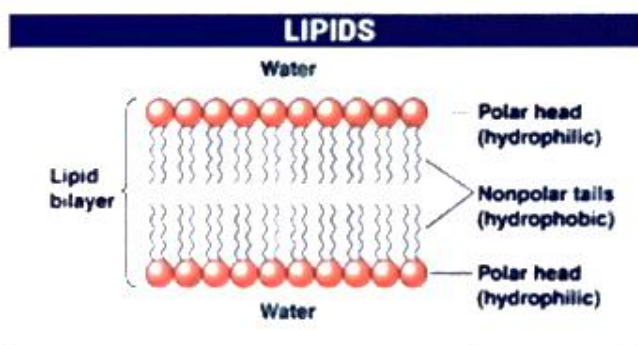
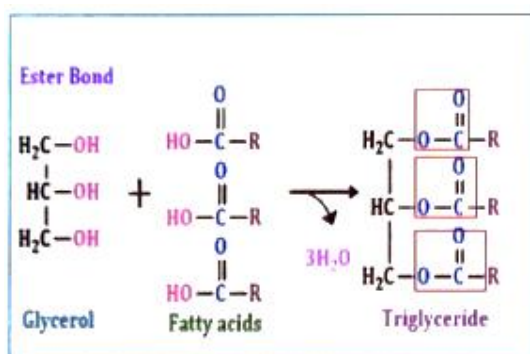
CARBOHYDRATE	BIOLOGICAL IMPORTANCE
D-Glucose	- It is carried by the blood and used by the tissue. - Its presence in the urine cause glycosuria.
D-Galactose	- Synthesized in mammary gland to make the lactose of milk. - It is constituent of glycolipid and glycoprotein.
D-Mannose	- It is constituent of prosthetic polysaccharides of albumin, glycoprotein and blood group.

CARBOHYDRATE	BIOLOGICAL IMPORTANCE
Agar	-Prepared from sea weeds and contain glucose . -It is used to culture bacterial colonies because does not digest it
Agarose	-It is made up of galactose combine with 3,6- anhydro galactose unit
Dextrose	-It is dextrorotatory glucose(D-glucose)
Dextrin	-Partially digested product of starch by amylase on diluted acid
Dextran	-Homopolysaccharide made up of glucose. -It is synthesized by bacteria , Antithrombotic , Volume expander
Inulin	-Polysaccharides containing fructofuranose unit it is used to measure GFR and renal clearance

LIPIDS

❖ DEFINITION TO LIPIDS

- Lipids are a group of compounds related to fatty acids and are **insoluble in water** but **soluble** in organic solvents.
- They are **esters of fatty acids with alcohol esters** and are utilizable by the living organism.



Bloor's criteria:

According to **Bloor**, lipids have the following **characteristics**:



1. They are insoluble in water.
2. **Soluble in organic solvents** like ether, chloroform, benzene, acetone.
3. Related to **fatty acids as esters**.
4. They are **utilised by living organisms**.



Role of lipids

1. Lipids are important **constituents of diet**.
2. They act as a **fuel and yield energy**.
3. They act as **insulation** and protect **vital organs**.
4. They provide **building blocks for high molecular weight substances** e.g. **acetic acid** is used for the synthesis of **cholesterol** and some **hormones**.
5. They supply **essential fatty acids** which are **not synthesized in the body**.
6. They act as **carriers for fat soluble vitamins**.
7. They are important **structural components of cell membrane**.

❖ CLASSIFICATION OF LIPIDS:

S.No	Class	Definition	Examples
1	Simple lipid	Ester of fatty acid with alcohol	✓ Fats ✓ Waxes
2	Complex Compound Lipid	They are esters of fatty acids with alcohols and contain additional groups (e.g. carbohydrates, phosphates, proteins etc).	✓ Phospholipids ✓ Glycolipids ✓ Sulfolipids ✓ Lipoproteins
3	Derived lipids	These are substances derived from the above groups by hydrolysis .	✓ Fatty acids ✓ Alcohols ✓ Sterols and steroids ✓ Polyisoprenoids

Lipids are classified as:

1. Simple lipids
2. Compound lipids
3. Derived lipids



1. Simple lipids:

They are esters of fatty acids with **alcohols**.

They are further classified as:

- (a) **Triglycerides or Fats** – esters of fatty acids with **glycerol**.
- (b) **Waxes** – esters of fatty acids with **higher alcohols**.



2. Complex (or compound) lipids :

- These are **esters of fatty acids with alcohols containing additional groups** such as phosphate, nitrogenous base, carbohydrate, protein etc. They are further divided as follows

(a) Phospholipids :

- They contain **phosphoric acid and frequently a nitrogenous base**.
- This is in addition to alcohol and fatty acids.

(i) **Glycerophospholipids** : These phospholipids contain glycerol as the alcohol e.g., lecithin, cephalin.

(ii) **Sphingophospholipids** : Sphingosine is the alcohol in this group of phospholipids e.g., sphingomyelin.

(b) Glycolipids :

- These lipids contain a **fatty acid, carbohydrate and nitrogenous base**.
- The alcohol is **sphingosine**, hence they are also called as glycosphingolipids.
- **Glycerol and phosphate are absent** e.g., cerebrosides, gangliosides.

(c) Lipoproteins :

- Macromolecular complexes of **lipids with proteins**.

(d) Other complex lipids :

- **Sulfolipids, aminolipids and lipopolysaccharides** are among the other complex lipids.

3. Derived lipids :

- These are the derivatives obtained on the **hydrolysis of group 1 and group 2 lipids** which possess the characteristics of lipids.

- These include **glycerol and other alcohols, fatty acids, mono- and diacylglycerol, lipid (fat) soluble vitamins, steroid hormones, hydrocarbons and ketone bodies.**
-

4. Miscellaneous lipids :

- These include a large number of compounds possessing the characteristics of lipids e.g., carotenoids, squalene, hydrocarbons such as pentacosane (in bees wax), terpenes etc.

NEUTRAL LIPIDS :

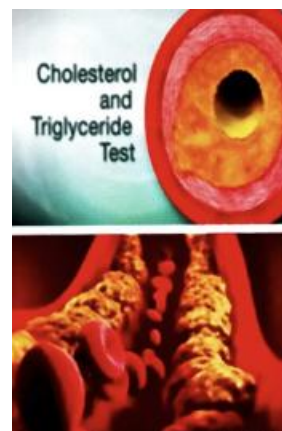
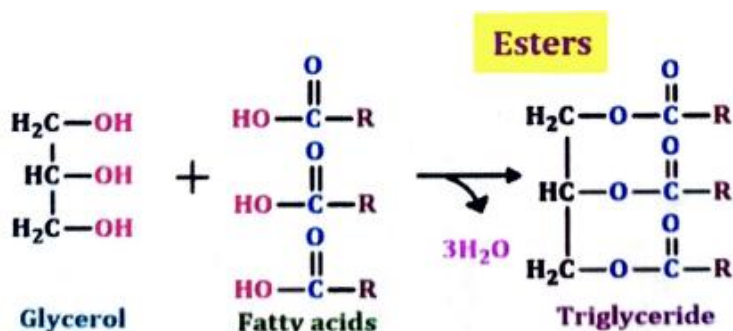
- The lipids which are uncharged are referred to as neutral lipids. These are mono-, di-, and triacylglycerols, cholesterol and cholesteryl esters.
-

❖ Biological Role of Lipids

- Lipids perform several important functions:
 1. They are the concentrated **fuel reserve of the body (triacylglycerols).**
 2. Lipids are the constituents of membrane **structure and regulate the membrane permeability (phospholipids and cholesterol).**
 3. They serve as a source of **fat soluble vitamins (A, D, E and K).**
 4. Lipids are important as **cellular metabolic regulators** (steroid hormones and prostaglandins).
 5. Lipids protect the **internal organs, serve as insulating materials** and give shape and smooth appearance to the body.
-

❖ Structure of Triglycerides (oils and fats)

- **Fat (triglyceride)** is formed by the **condensation** of one molecule of glycerol with three molecules of fatty acids.



❖ Properties of triglycerides (oils and fats)

Physical properties :

1. Fats are **insoluble in water**. But they are **soluble in organic solvents** like benzene, ether and chloroform.
2. They have **no colour, odour or taste**.
3. They have a **low melting point**.
4. They have a **low specific gravity**. So they float in water.

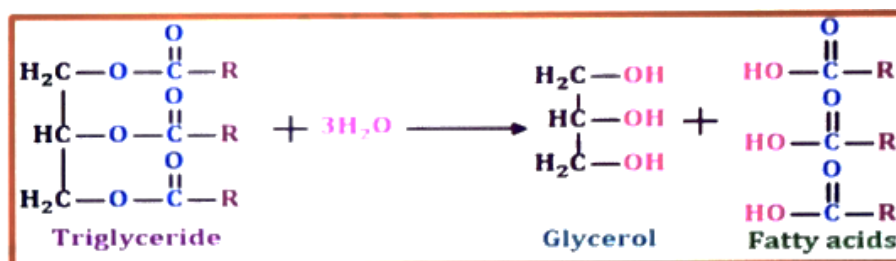


Chemical properties :

1. Hydrolysis:

- Fats can be **hydrolysed by acids**, alkalies, steam or enzymes called **lipases**.

- They act on the **ester linkage** and liberate **glycerol** and **fatty acid**.

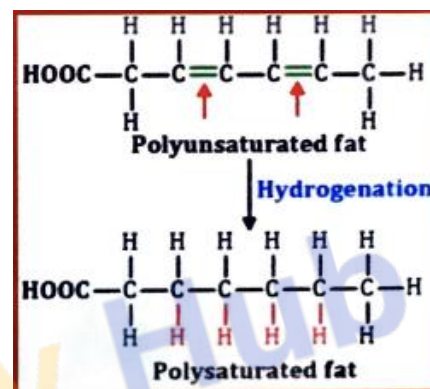


2. Saponification:

- Hydrolysis of a **fat** by **alkali** is called as **saponification**.
- The products of hydrolysis are **glycerol** and **alkali salts of fatty acids**.
- The alkali salts of fatty acids are called as **soaps**.
- They are insoluble but their **sodium** and **potassium** salts are soluble.

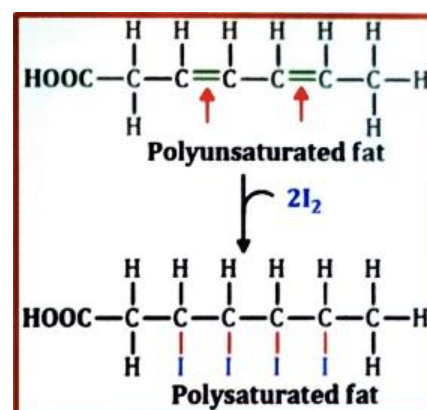
3. Hydrogenation:

- Unsaturated fats combine with **hydrogen** in presence of a catalyst like **nickel**.
- Hydrogen is added at the double bonds of unsaturated fats and they are converted into **saturated fats**.



4. Halogenation :

- Unsaturated fats **combine with halogens** like **I₂** or **Br₂** at the double bonds.
- Two halogen atoms are taken up for **each double bond** as follows:



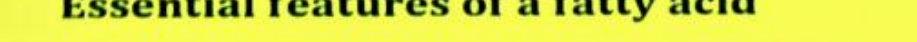
5. Rancidity :

- It is the **chemical change** of a fat resulting in an **unpleasant odour and taste**.
- Rancidity occurs when fats are **exposed to light, heat, air or moisture**.

-
- OXIDATIVE STRESS**
- Normal cell Cell attacked by free radicals Cell with oxidative stress

- Fatty acid are **carboxylic acid** with hydrogen chain (**-CH₂-CH₂-CH₂-**) and represented by a chemical formula **R-COOH**, where R stands for hydrocarbon chain.
- The fatty acid are **amphipathic** in nature i.e. each has **hydrophilic** (COOH) and **hydrophobic** (hydrocarbon chain) group in the structure.

Essential features of a fatty acid


The diagram illustrates the chemical structure of a saturated fatty acid. It features a long, straight hydrocarbon chain composed of 17 carbon atoms. Each carbon atom in the chain is bonded to hydrogen atoms, with the terminal carbons having three hydrogens and the internal carbons having two. The chain is terminated by a carboxylic acid group, which consists of a carbon atom double-bonded to an oxygen atom and single-bonded to a hydroxyl (-OH) group.

long hydrocarbon chain

carboxylic acid group

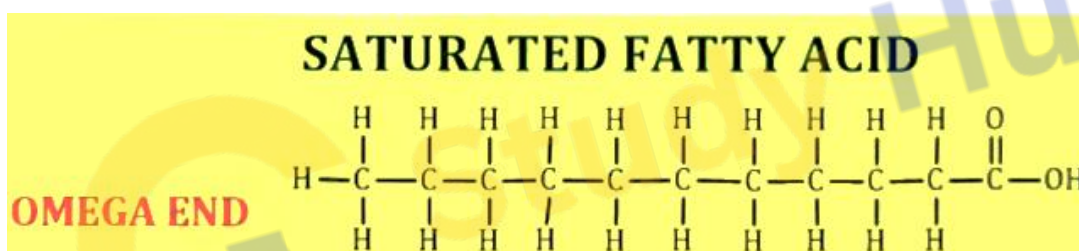
◆ Classification based on chemical and nutritional requirements

SATURATED FATTY ACID	UNSATURATED FATTY ACID
- Acetic acid (2C) - Propionic acid (3C) - Valeric acid (5C) - Caproic acid (6C) - Lauric acid (12C) - Myristic acid (14C) - Palmitic acid (16C) - Stearic acid (18C) - Arachidic acid (20C)	- Palmitoleic acid (16-C, 1 Double bond) - Oleic acid (18-C, 1 Double bond) - Linoleic acid (18-C, 2 Double bond) - Linolenic acid (18-C, 3 Double bond) - Arachidonic acid (20-C, 4 Double bond)
Essential fatty acid: - Fatty acid that cannot be synthesized by human body, commonly they are polyunsaturated fatty acid (PUFA). - Arachidonic acid - Linoleic acid - Linolenic acid	

Saturated fatty acids

- Saturated fatty acids **do not contain double bonds**.

Examples:

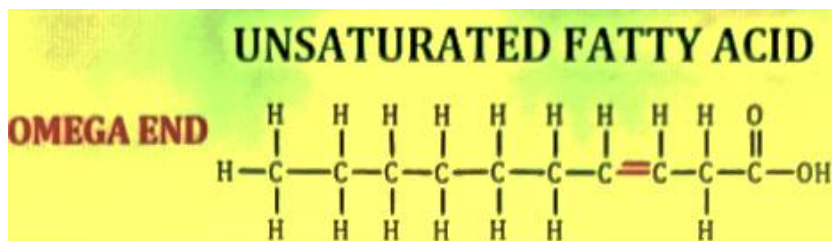


Fatty acid	No. of carbon	Structure	Fatty acid	No. of carbon	Structure
Butyric acid	4	$\text{CH}_3(\text{CH}_2)_2\text{COOH}$	Lauric acid	12	$\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$
Caproic acid	6	$\text{CH}_3(\text{CH}_2)_4\text{COOH}$	Myristic acid	14	$\text{CH}_3(\text{CH}_2)_{12}\text{COOH}$
Caprylic acid	8	$\text{CH}_3(\text{CH}_2)_6\text{COOH}$	Palmitic acid	16	$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$
Capric acid	10	$\text{CH}_3(\text{CH}_2)_8\text{COOH}$	Stearic acid	18	$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$

Unsaturated fatty acids

- Unsaturated fatty acids **contain one or more double bonds**.

Examples:



Name and Source	No. of carbon	Structure
Palmitoleic acid (Cotton seed oil)	16	$\text{CH}_3(\text{CH}_2)_5\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$
Oleic acid (Safflower oil)	18	$\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$
Linoleic acid (Sesame oil, Sunflower oil)	18	$\text{CH}_3(\text{CH}_2)_4\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$
Linolenic acid (Linseed, Soyabean oil)	18	$\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$
Arachidonic acid	20	$\text{CH}_3(\text{CH}_2)_4(\text{CH}=\text{CHCH}_2)_4(\text{CH}_2)_2\text{COOH}$

Phospholipids

- Phospholipids are complex lipids **containing phosphoric acid**.
- Glycerophospholipids contain glycerol as the alcohol and these include **lecithin, cephalin, phosphatidylinositol, plasmalogen and cardiolipin**.

Sphingophospholipids

- (Sphingomyelins) contain sphingosine as the alcohol in place of glycerol (in glycerophospholipids).
- Phospholipids are the major constituents of plasma membranes.

Cerebrosides

- Cerebrosides are the **simplest form of glycolipids** which occur in the membranes of nervous tissue.
- Gangliosides are predominantly found in the ganglions.
- They contain one or more molecules of N-acetylneuraminic acid (**NANA**).

Steroids

- Steroids contain the ring **cyclopentanoperhydrophenanthrene**.
- The steroids of biological importance include cholesterol, bile acids, vitamin D, sex hormones and cortical hormones.
- A steroid containing one or more hydroxyl groups is known as **sterol**.

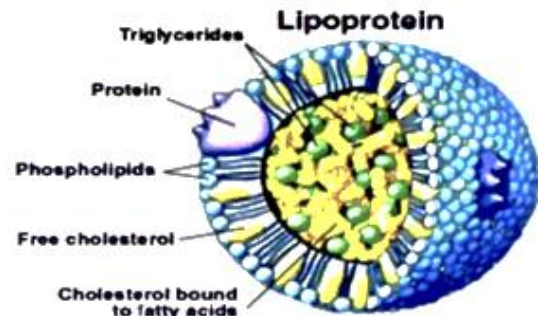
Cholesterol

- Cholesterol is the most abundant animal sterol.
- It contains one hydroxyl group (at **C₃**), a double bond (**C₅-C₆**) and an eight carbon side chain attached to **C-17**.
- Cholesterol is a constituent of membrane structure and is involved in the **synthesis of bile acids, hormones (sex and cortical) and vitamin D**.

The lipids that possess **both hydrophobic (non -polar) and hydrophilic (polar) groups are known as amphipathic**. These include fatty acids, phospholipids, sphingolipids and bile salts. Amphipathic lipids are **important constituents in the bilayers of the biological membranes**.

❖ Lipoproteins

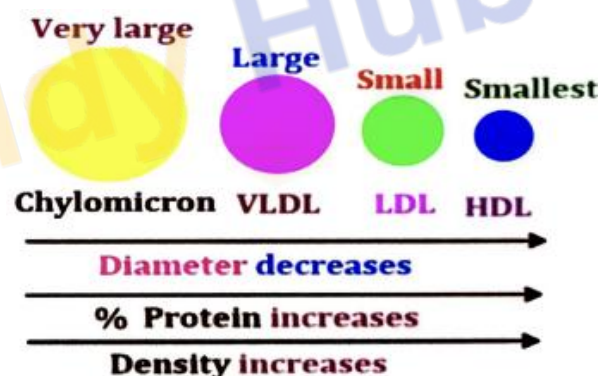
- Lipoproteins are **large water-soluble complexes** formed by a combination of lipid and **protein** that transport insoluble lipids through the **blood between different organs and tissues**.
- Lipoproteins consist of a lipid core containing nonpolar **triacylglycerol** and **cholesterol ester** surrounded by a single layer of amphipathic phospholipids and free **cholesterol molecules** with some proteins (apoprotein).



❖ Classes of lipoproteins

- Lipoproteins have been categorized into **four major classes** according to their **physical and chemical properties**.

1. **Chylomicrons**
2. **Very low density lipoproteins (VLDL)**
3. **Low density lipoproteins (LDL)**
4. **High density lipoproteins (HDL)**

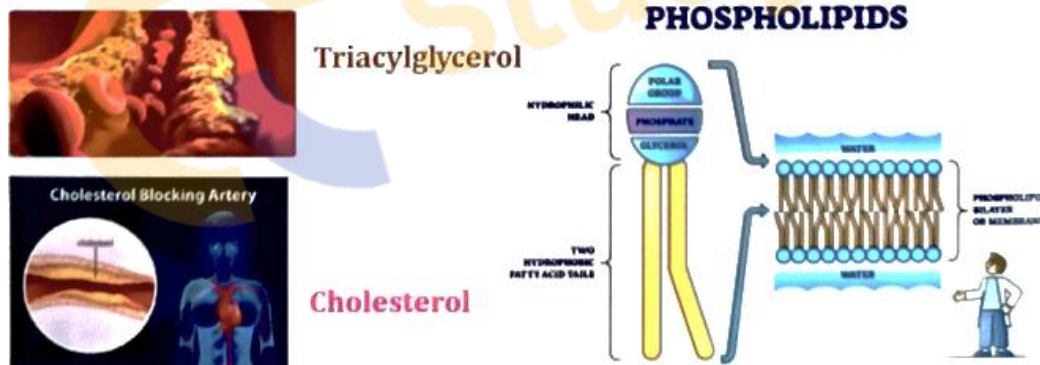


- The **density** of these **lipoproteins** is **inversely proportional** to **triacylglycerol** content.
- As the **density increases**, the diameter of the **particle decreases**.
- Chylomicrons** containing about **1 percent protein** and **99 percent triacylglycerol** have the lowest density.
- While **HDL** containing **50 percent protein** and **50 percent lipid** have the highest density.
- Triacylglycerol** is the predominant lipid in chylomicrons and VLDL.

- **Cholesterol** is the predominant lipid in LDL, whereas **phospholipids** are the predominant lipid in HDL.

❖ Characteristics of human plasma lipoproteins

S.NO.	Variable	Chylomicron	VLDL	LDL	HDL
1.	Diameter (nm)	70 to 1200	25 to 70	20 to 25	5 to 10
2.	Lipid-Protein ratio	99:1	90:10	80:20	50:50
3.	Major lipids	Triacylglycerol	Triacylglycerol	Cholesterol	Phospholipids
4.	Lipid components (%)				
	Triacylglycerol	86	55	6	5
	Cholesterol (free and ester)	5	20	50	20
	Phospholipid	7	18	22	25



❖ The four main lipoproteins and their site of synthesis and function

S.No	Lipoprotein	Site of Synthesis	Function
1.	Chylomicrons	Intestine	Transport of dietary lipids from intestine to peripheral tissues
2.	VLDL	Liver	Transport of triacylglycerol from liver to peripheral tissues
3.	LDL	Plasma, VLDL	Transport of cholesterol from liver to peripheral tissues
4.	HDL	Liver and intestine	Transport of free cholesterol from peripheral tissues to the liver (Reverse cholesterol transport)

◆ Test for lipids

S.No	Test	Observation	Inference
1.	Salkowski test (Concentrated H ₂ SO ₄ and chloroform)	Upper layer (chloroform) → changes to Red, Lower layer (H ₂ SO ₄) → changes to Yellow	Test for Cholesterol
2.	Liebermann Burchard (2-3 ml of organic solution + 5-6 drop acetic anhydrous + H ₂ SO ₄)	Rose to bluish green colour solution	Test for Cholesterol
3.	Grease spot test (oil and water)	Some oil and water are smeared into paper after some time water smear would become translucent but smear of oil is translucent.	Test for Fat

S.No	Test	Observation	Inference
4.	Sudan IV test (Sudan dye)	Sudan red is the fat-soluble dye that stains lipid red.	Test for Fat

UNIT-1: BIOMOLECULES-PART-2

Points to be covered in this topic

- ❖ **Nucleic Acid**
 - Introduction
 - Classification
 - Chemical nature
 - Biological role
- ❖ **Protein & Amino acids**
 - Introduction
 - Classification
 - Chemical nature
 - Biological role

Nucleic Acids

- Nucleic acids are polymers of **nucleotides**, linked by **phosphodiester bond**, they are therefore called **polynucleotides**.
- **The nucleic acids are of two main types:**

- **Deoxyribonucleic acid** or DNA
- **Ribonucleic acid** or RNA
- DNA is present in nuclei and small amounts are also present in mitochondria, whereas **90% of the RNA** is present in cell cytoplasm and **10% in the nucleolus**.

❖ Nitrogen bases of RNA & DNA

- Two classes of nitrogenous bases namely **purines** and **pyrimidines** are present in RNA and DNA.

Purine bases

- A purine is a heterocyclic aromatic organic compound, consisting of a pyrimidine ring fused to an imidazole ring.
- **Two principal** purine bases found in DNAs, as well as **RNAs** are:

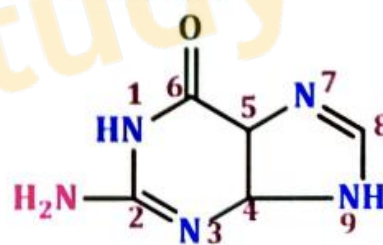


i. Adenine (A)



Adenine (6-aminopurine)

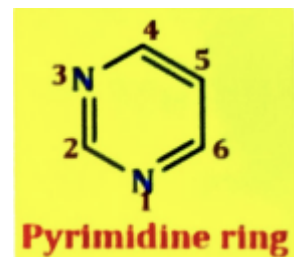
ii. Guanine (G)



Guanine (2-amino-6-oxypurine)

Pyrimidine bases

- Pyrimidine is a **heterocyclic aromatic organic compound** similar to **benzene** and **pyridine**, containing two nitrogen atoms at **positions** 1 and 3 of the six-member ring.
- Three major pyrimidine bases are:



i. Cytosine (C)

ii. Uracil (U)

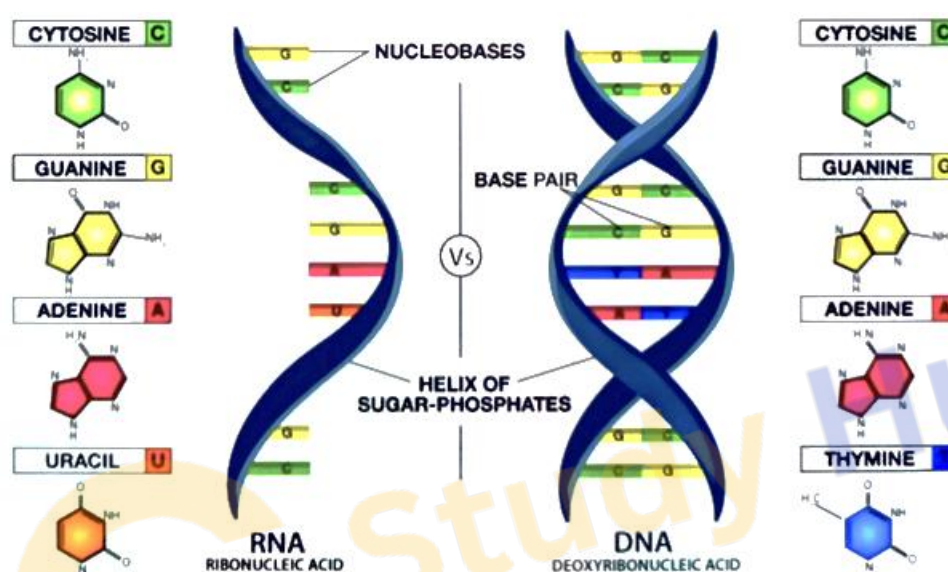
iii. Thymine (T)



- **Cytosine** and **uracil** are found in RNAs and cytosine and thymine in DNA.
- Both DNA and RNA contain the pyrimidine base **cytosine** but they differ in their second pyrimidine base.
- DNA contains **thymine** whereas RNA contains **uracil**.

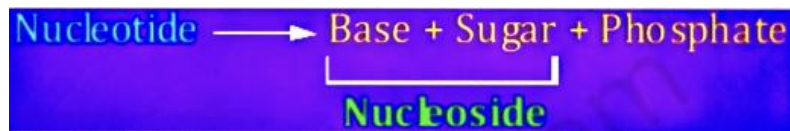


❖ Nitrogen bases of RNA and DNA



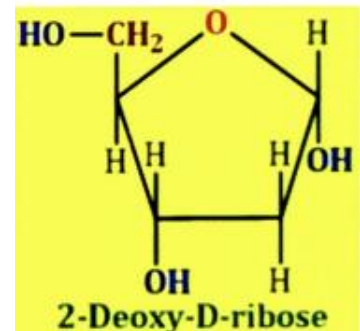
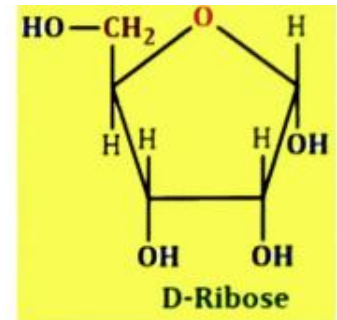
❖ Nucleosides and Nucleotides

- **Nucleosides are composed of:**
 - Nitrogenous base (purine/pyrimidine)
 - Pentose sugar (ribose/deoxyribose)
- **Nucleotides are composed of:**
 - Nitrogenous base (purine/pyrimidine)
 - Pentose sugar (ribose/deoxyribose)
 - Phosphate group



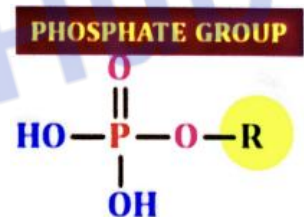
✓ PENTOSE SUGAR (RIBOSE/DEOXYRIBOSE)

- The **pentose sugar** is either **D-ribose** or **D-2-deoxyribose**.
- DNA and RNA are distinguished on the basis of the pentose sugar present.
 - DNA contains **D-2-deoxyribose**
 - RNA contains **D-ribose**
- A **pentose sugar** (**D-ribose** or **D-2-deoxyribose**) is linked to a **base** (purine or pyrimidine) via **covalent N-glycosidic bond**.



✓ PHOSPHATE GROUP

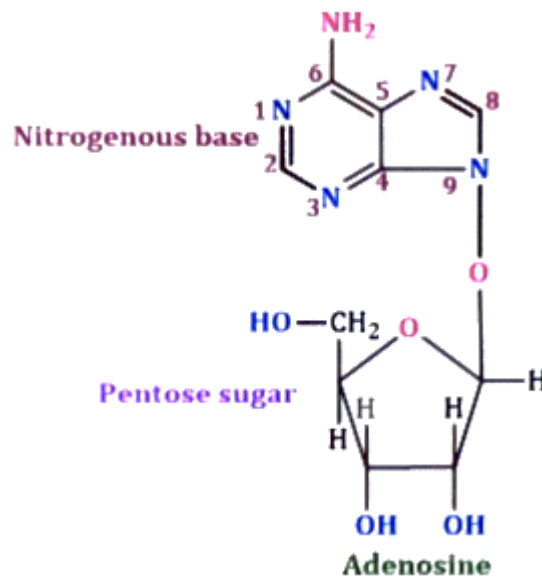
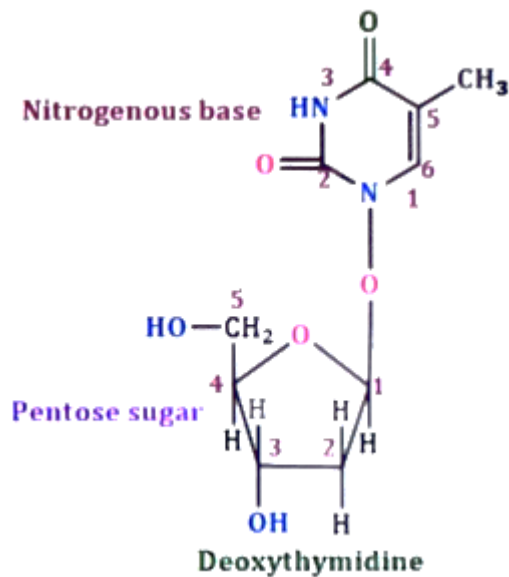
- Mononucleotides** are nucleosides in which a **single phosphate group** is attached to the hydroxyl group of the pentose sugar.
 - For example, **AMP** (**adenosine monophosphate**) is **adenine + ribose + phosphate**.
- If an additional **phosphate group** is attached to the pre-existing phosphate of mononucleotide:
 - A **nucleoside diphosphate**, e.g., **ADP**
 - A **nucleoside triphosphate**, e.g., **ATP**



Nucleosides

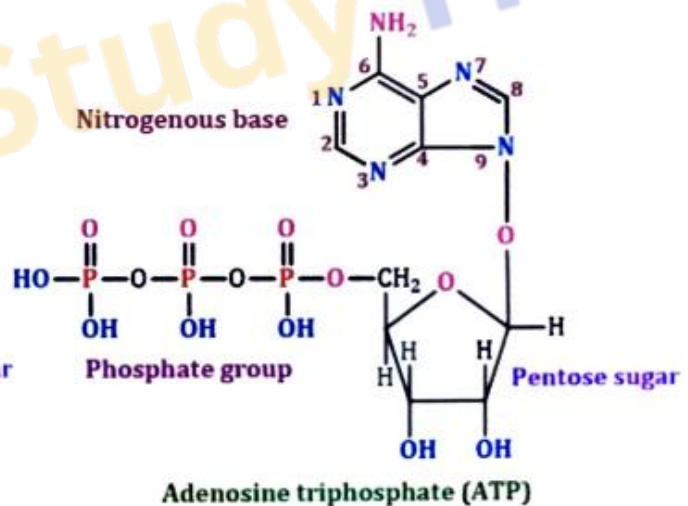
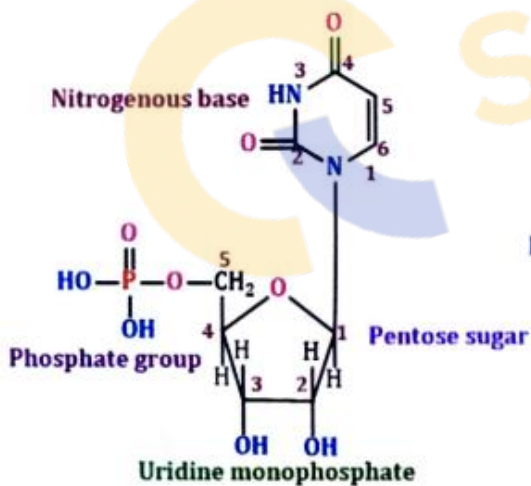
- A **nucleoside** consists of a **nitrogenous base** (purines and pyrimidines) and **pentose sugar** (ribose or deoxyribose) by the **β -N-glycosidic bond** between

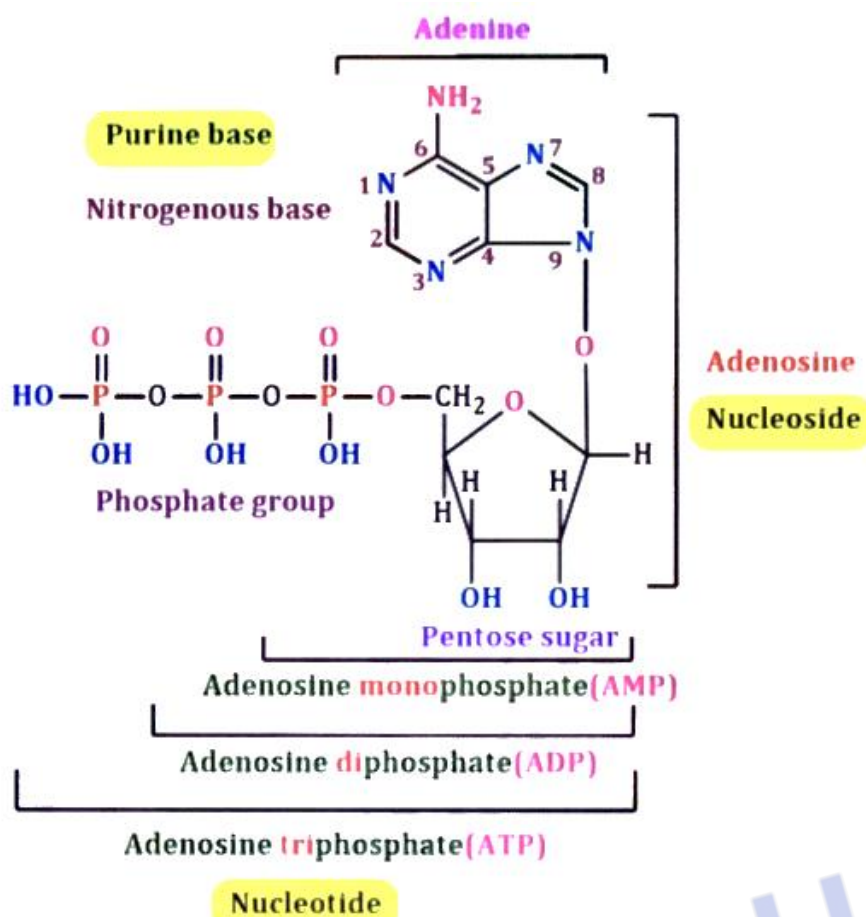
the **first carbon** of the pentose sugar and **N9** of a purine or **N1** of a pyrimidine.



Nucleotides

- A **nucleotide** consists of a **nitrogenous base** (purines and pyrimidines), **pentose sugar** (ribose or deoxyribose), and **phosphate group**.





❖ Differentiation between Nucleosides and Nucleotides

	Nucleotide	Nucleoside
DEFINITION	Nucleotide is a monomer of nucleic acids like DNA and RNA .	Nucleoside is a nucleobase linked to a pentose sugar molecule.
COMPONENTS	A pentose sugar , a nitrogenous base and a phosphate group.	A nitrogenous base and a pentose sugar .
FUNCTIONS	They polymerize and form nucleic acids. As well as they act as energy-storing molecules.	Nucleosides are good anticancer compounds. As well as they have antiviral properties .

EXAMPLES	Deoxyribonucleotides, ribonucleotides, ATP, NADP, FAD, CAM, etc.	Examples of nucleosides include cytidine, uridine, adenosine, guanosine, thymidine and inosine.
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NUCLEIC ACID

- **Nucleic acid** serves as a transmitter of **genetic information**.
- Two types of nucleic acid:
 1. **DNA** (Deoxyribonucleic acid)
 2. **RNA** (Ribonucleic acid)

❖ History

DNA was discovered by	Friedrich Miescher
Information of DNA containing genetic material	Avery, Macleod and Mac Carty.
Double helical structure of DNA was proposed by	D-Watson and Crick

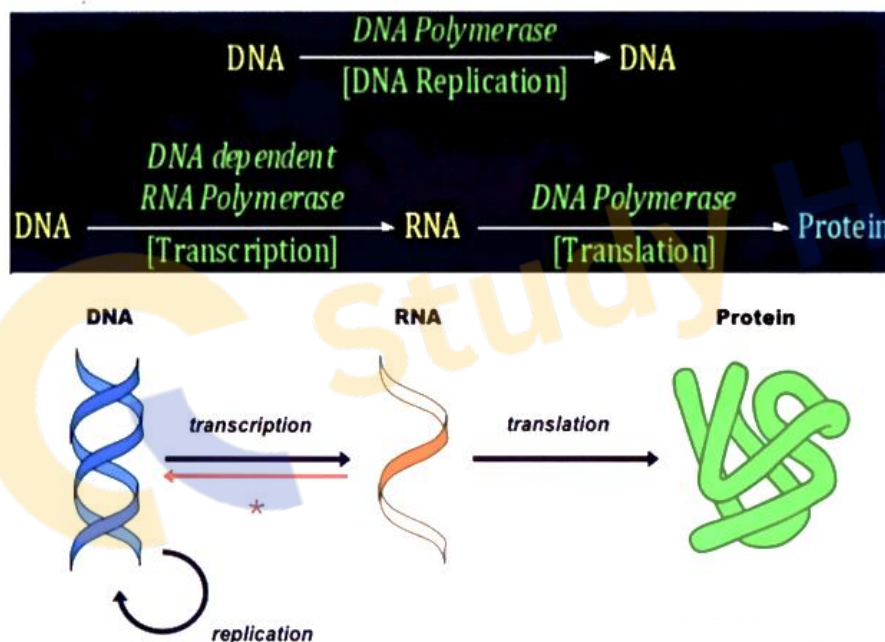
❖ DNA & RNA

- **Deoxyribonucleic acid (DNA)**
 - Basic unit of **genetic information**.
 - **DNA** is a polymer of **deoxyribonucleotides**.
 - **Double helical structure** of DNA was proposed by **Watson and Crick**.
 - **Dextrorotatory** (polarized light to right).
- **Ribonucleic acid (RNA)**

- RNAs are **single stranded**.
- RNA is an **unbranched** linear polymer of **ribonucleotides** joined by 3', 5' phosphodiester bonds.
- The **phosphodiester bonds** join the **3'-OH group** of ribose of one nucleotide with the **5'-OH group** of ribose sugar of the next nucleotide.

❖ Central Dogma of Life

- The interrelationship of these three biomolecules (**DNA**, **RNA** and **Proteins**) is called **central dogma of molecular biology**.



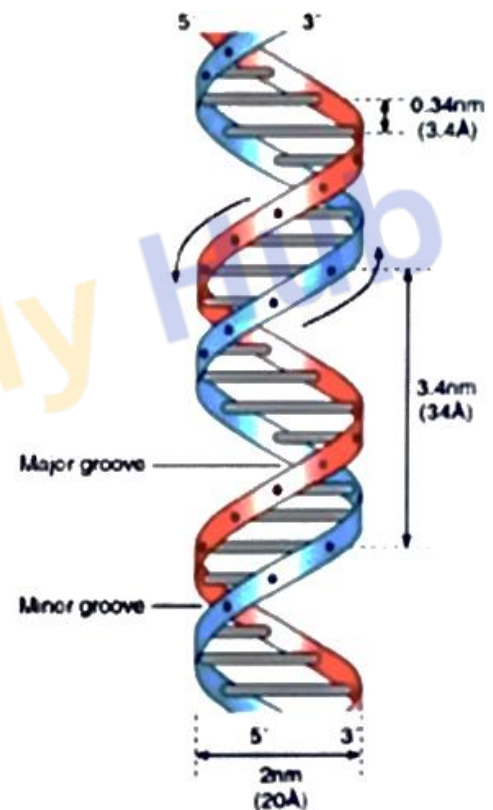
❖ Watson-Crick DNA Double Helical Structure

- In **1953**, **James Watson** and **Francis Crick** deduced the **three-dimensional** structure of DNA.



The Important Features of Their Model of DNA Are As Follows:

- **Two helical polynucleotide chains** are coiled around a common axis. The chains run in opposite directions (**anti parallel**).
- The **purine and pyrimidine bases** are on the inside of the **helix**, whereas the **phosphate and deoxyribose units** are on the outside.
- The **diameter of the helix** is **20 Å**. Adjacent bases are separated by **3.4 Å** along the helix axis and the helical structure repeats after ten residues on each chain, i.e., at **intervals of 34 Å**.
- The **two chains** are held together by **hydrogen bonds** between complementary pairs of bases:
 - **Adenine** is always paired with **thymine** by formation of **two hydrogen bonds**.
 - **Guanine** is always paired with **cytosine** by formation of **three hydrogen bonds**.
- The **two strands** are always complementary to each other. In a double stranded DNA molecule, the content of **adenine** equals that of **thymine** and the contents of **guanine** equals that of **cytosine**.
- The model proposed by **Watson and Crick** is a **B form of DNA (B-DNA)** which is a **right handed helix of 10 base pairs per turn**, containing grooves of alternate sizes, known as **major and minor grooves**.
- Other forms of **DNA** may also occur, such as **A-DNA** and **Z-DNA**. Under physiologic conditions, DNA is almost entirely in **Watson-Crick B form**.

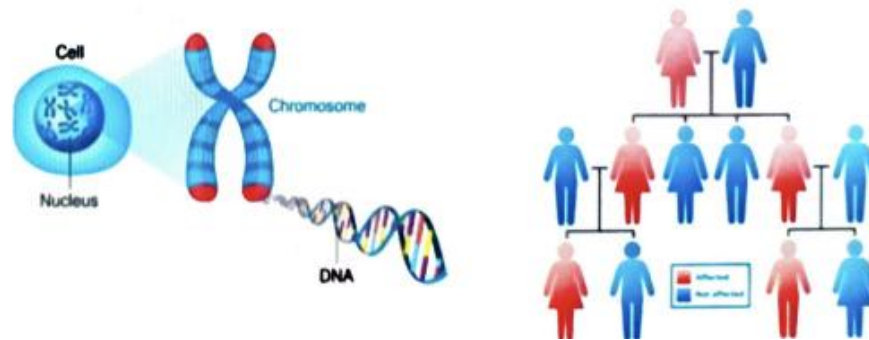


TYPES OF DNA

FEATURE	B-DNA	A-DNA	Z-DNA
Helix type	Right-handed	Right-handed	Left-handed
Helical Diameter (nm)	2.37	2.55	1.84
Distance. Per each complete turn (nm)	3.4	3.2	4.5
Rise per base pair (nm)	0.34	0.29	0.37
Number of base pairs per complete turn	10	11	12
Base pair tilt	+19°	-1.2° (variable)	-9°
Helix axis rotation	Major groove	Through base pairs (variable)	Minor groove

BIOLOGICAL FUNCTIONS OF NUCLEIC ACIDS:-

- DNA is exclusively responsible for maintaining the **identity of different species of organisms over millions of years.**
- DNA is the chemical basis of **heredity** and may be regarded as the reserve of **genetic information.**
- A DNA molecule is capable of **self-duplication during cell division and identical DNA strands are transferred to daughter cells.**
- Another important function of nucleic acids is the **protein synthesis in the cell.**
- Actually, the proteins are synthesised by various RNA molecules in the cell but the message for the **synthesis of a particular protein is present in DNA.**

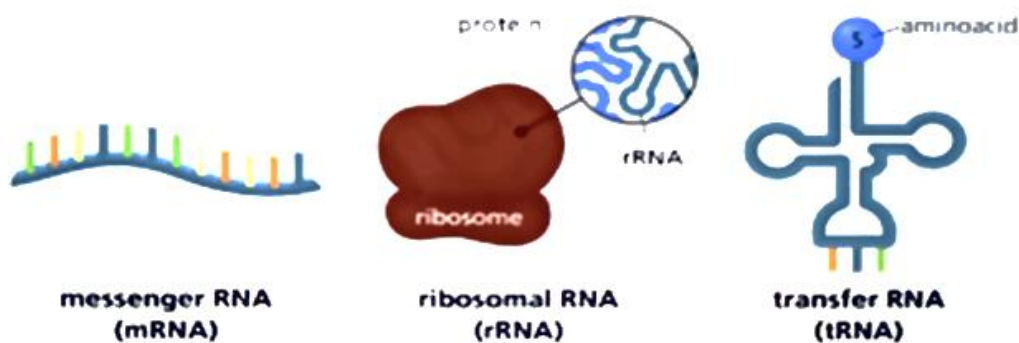


❖ Ribonucleic acid (RNA)

- RNAs are **single stranded**.
- RNA is an **unbranched linear polymer of ribonucleotides** joined by 3', 5' phosphodiester bonds.
- The phosphodiester bonds join the **3'-OH group** of ribose of one nucleotide unit to the **5'-OH group** of ribose sugar of the next nucleotide.

TYPES OF RNA

- Cell contains **three major types of RNA**:
 1. Messenger RNA (mRNA)
 2. Transfer RNA (tRNA)
 3. Ribosomal RNA (rRNA)
- All of these are involved in the process of **protein biosynthesis**. Each differs from the others by **size and function**.



TYPE OF RNA

TYPE OF RNA	ABBREVIATION	FUNCTION
Messenger RNA	mRNA	Transfer genetic information from genes to ribosomes to synthesize protein
Heterogeneous nuclear RNA	hnRNA	Serve as precursor for mRNA & other protein
Transfer RNA	tRNA	Transfer amino acid to mRNA for protein biosynthesis

TYPE OF RNA	ABBREVIATION	FUNCTION
Ribosomal RNA	rRNA	Provide structural frame work for ribosome
Small nuclear RNA	snRNA	Involve in mRNA processing
Small nucleolar RNA	snoRNA	Involve role in processing of rRNA
Transfer messenger RNA	tmRNA	Mostly presence in bacteria add short peptide tag to protein to facilitate the degradation of incorrectly Protein synthesis

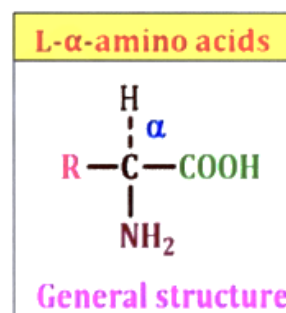
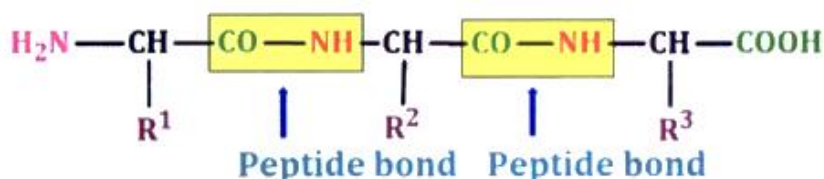
❖ Difference between DNA and RNA

CHARACTER	DNA	RNA
Presence	In chromosome & little in mitochondria & chloroplast	Distribution throughout the cytoplasm except vacuoles
Strands	Double	Single
Sugar	Deoxyribose sugar	Ribose sugar
Nitrogenous base	A, G, C & T	A, G, C & U
Transcription	DNA → DNA (Replication) RNA → RNA (transcription)	RNA does not replicate
Nucleotide	Made up to large no. of nucleotide up to 4 million	Consist of less no. of nucleotide up to 12000
Stain	It is fuelgen (+ve)	It is fuelgen (-ve)

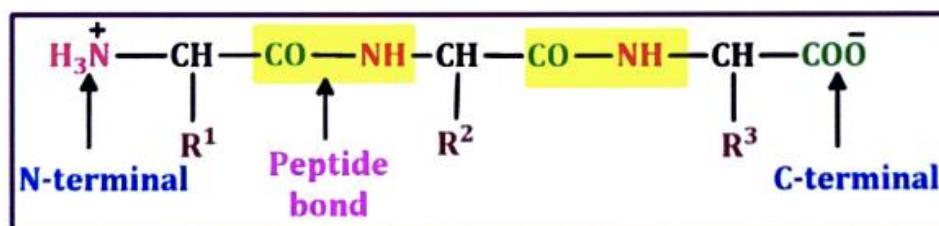
Amino acids and Proteins

❖ DEFINITION TO PROTEINS

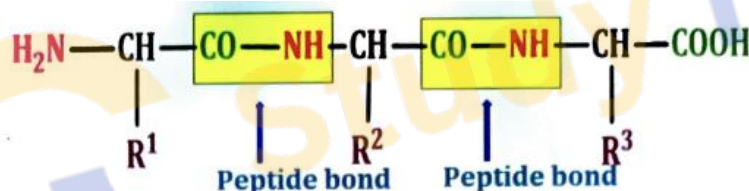
- **Proteins are nitrogen containing**, most abundant **organic macromolecules** widely distributed in animals and plants. They perform **structural and dynamic functions in the organisms**.
- Proteins are **polymers composed of L-α-amino acids** join by the **peptide bond**.
- Proteins are **essential nutrients for building block of body tissue** and proteins contain **4 Kcal gram as fuel**.



- All naturally occurring **amino acids** are **levorotatory except glycine** which is optically inactive (No chiral carbon).
- **N-terminal end** – The free **NH₂ group of the terminal amino acid** is called as N-terminal end.
- **C-terminal end** – The free **COOH end** is called **C-terminal end**.



- In proteins, **amino acids** are joined covalently by **peptide bonds**, which are formed between **α -carboxyl group** of one amino acid and **α -amino group** of another with the **elimination of a water molecule**.
- **Linkage of many amino acids** through peptide bonds results in an **unbranched chain called a polypeptide**.



❖ PROTEIN CLASSIFICATION BASED ON CHEMICAL NATURE AND SOLUBILITY

- This is a more comprehensive and popular classification of proteins. It is based on the amino acid composition, structure, shape and solubility properties. Proteins are broadly

CLASSIFIED INTO 3 MAJOR GROUPS

1. **Simple proteins:** They are composed of only **amino acid residues**.

2. **Conjugated proteins:** Besides the amino acids, these proteins **contain a non-protein moiety known as prosthetic group or conjugating group.**
3. **Derived proteins:** These are the **denatured or degraded products** of simple and conjugated proteins.

S.NO.	CLASS	SUB-CLASS	EXAMPLES
1.	Simple	Globular protein	Albumins, Globulin, Glutelin, Prolamin, Histone, Globin, Protamine, Lectin.
		Scleroprotein	Collagen, Elastin, Keratin.
2.	Conjugated		Nucleoprotein, Glycoprotein, Mucoprotein, Lipoprotein, Phosphoprotein, Chromoprotein, Metalloprotein.
3.	Derived	Primary	Coagulated protein, Protean, Metaprotein.
		Secondary	Proteose, Peptone, Polypeptide, Peptide.

❖ Classification of proteins on the basis of reactivity:

1. SIMPLE PROTEINS

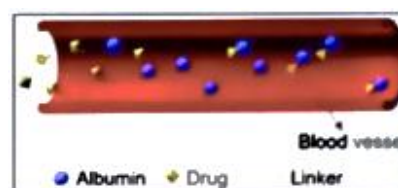
(a) Globular proteins:

- These are **spherical or oval in shape, soluble in water or other solvents and digestible.**



(i) Albumins:

- **Soluble in water and dilute salt solutions** and coagulated by heat.
- e.g. serum albumin, ovalbumin (egg), lactalbumin (milk).



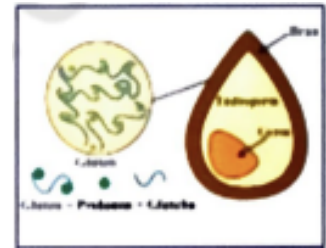
(ii) Globulins:



- Soluble in neutral and dilute salt solutions.
- e.g. serum globulins, vitelline (egg yolk).

(iii) Glutelins:

- Soluble in dilute acids and alkalies and mostly found in plants.
- e.g. Glutelin (wheat), Oryzenin (rice).



(iv) Prolamin:

- Soluble in 70% alcohol.
- e.g. Gliadin (wheat), Zein (maize).

(v) Histones:

- Strongly basic proteins, soluble in water and dilute acids but insoluble in dilute ammonium hydroxide.
- e.g. thymus histones.

(vi) Globin:

- These are generally considered along with histones.
- However, **globins are not basic proteins and are not precipitated by NH_4OH .**

(vii) Protamines:

- They are strongly basic and resemble histones but **smaller in size and soluble in NH_4OH .**
- Protamines are also found in association with **nucleic acids.**
- e.g. Sperm proteins.

(viii) Lectins:

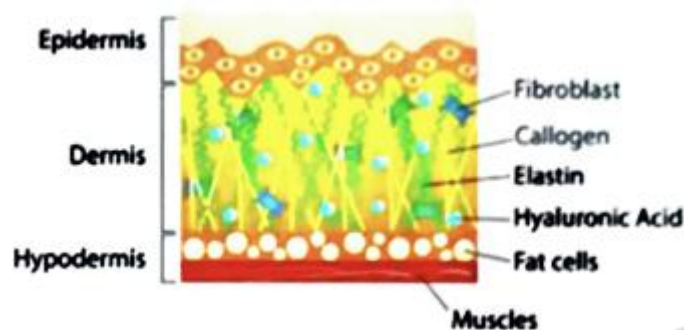
- **Lectins are carbohydrate-binding proteins**, and are involved in the interaction between cells and proteins.

- In the laboratory, lectins are useful for the **purification of carbohydrates** by affinity chromatography.
- e.g. **concanavalin A**, **agglutinin**.

(b) Fibrous proteins :

- These are **fiber like in shape**, **insoluble in water** and **resistant to digestion**. **Albuminoids or scleroproteins** are predominant group of fibrous proteins.

(i) **Collagens** are connective tissue proteins lacking tryptophan. Collagens, on boiling with water or dilute acids, yield gelatin which is soluble and digestible.



(ii) **Elastin** : These proteins are found in elastic tissues such as **tendons and arteries**.

(iii) **Keratins** : These are present in **exoskeletal structures** e.g. **hair, nails, horns**. Human hair keratin contains as much as **14% cysteine**.

2. Conjugated proteins

(a) Nucleoproteins :

- Nucleic acid (**DNA or RNA**) is the prosthetic group e.g. **Nucleohistones**, **Nucleoprotamines**.

(b) Glycoproteins :

- The prosthetic group is carbohydrate, which is less than 4% of protein. The term **mucoprotein** is used if the carbohydrate content is **more than 4%**. e.g. **Mucin (saliva)**, **Ovomucoid (egg white)**.

(c) Lipoproteins :

- Protein found in **combination with lipids** as the prosthetic group e.g. **serum lipoproteins**.

(d) Phosphoprotein :

- **Phosphoric acid** is the prosthetic group e.g. **casein (milk)**, **vitelline (egg yolk)**.

(e) Chromoprotein :

- The prosthetic group is **colored** in nature e.g. **Hemoglobin, Cytochromes**.

(f) Metalloprotein :

- These proteins contain **metal ions** such as **Fe, Co, Zn, Cu, Mg etc.** e.g. **Ceruloplasmin (Cu), carbonic anhydrase (Zn)**.

3. Derived proteins :

- The derived proteins are of **two types**.
- The **primary derived** are the **denatured or coagulated or first hydrolysed** products of proteins.
- The **secondary derived** are the **degraded** (due to breakdown of peptide bonds) products of proteins.

(a) Primary derived proteins

(i) Coagulated proteins :

- These are the **denatured proteins** produced by agents such as **heat, acids, alkalies** etc. e.g. **cooked proteins, coagulated albumin (egg white)**.

(ii) Proteans :

- These are the earliest products of **protein hydrolysis** by enzymes, dilute acids, alkalies etc. which are **insoluble in water**. e.g. **fibrin formed from fibrinogen**.

(iii) Metaprotein :

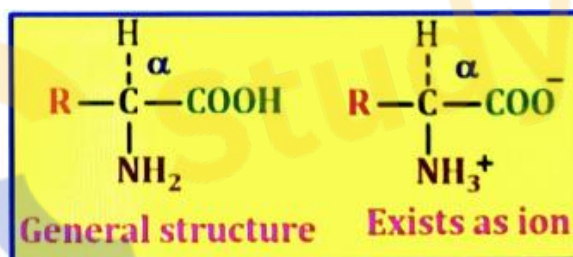
- These are the **second stage products** of protein hydrolysis obtained by treatment with slightly stronger acids and alkalies. e.g. **acid and alkali metaprotein**.

(b) Secondary derived proteins :

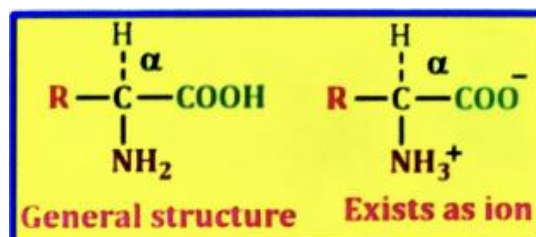
- These are the **progressive hydrolytic products** of protein hydrolysis. These include **proteoses, peptones, polypeptides and peptides**.

❖ DEFINITION OF AMINO ACID

- Amino acids are a group of **organic compounds** containing two functional groups—**amino and carboxyl**.
- The **amino group** (—NH_2) is **basic** while the **carboxyl group** (—COOH) is **acidic** in nature.



- Out of known 300, only 20 are known as standard amino acids.
- All amino acid have **L- configuration** & hence **L - α- amino acid**.
- All **naturally** occurring amino acids are optically active except glycine.
- The amino acid **carry equal number of positive and negative charge** i.e. no net charge and exist as **ZWITTER ION (AMPHOLYTIC)**.
- 21st amino acid** is the **Selenocysteine** precursor for the selenocysteine is serine.
- 22nd amino acid** is the **Pyrrolysine**.



❖ **Classification of amino acid based on chemical nature**

S. NO.	CATEGORY	EXAMPLE
1.	Neutral amino acid	Glycine, Alanine, Valine, Leucine, Isoleucine
2.	Acidic amino acid	Aspartic acid, Asparagine, Glutamic acid, Glutamine
3.	Basic amino acid	Arginine, Lysine
4.	Sulphur amino acid	Cysteine, Methionine
5.	Hydroxyl containing amino acid	Serine, Threonine
6.	Aromatic amino acid	Phenylalanine, Tyrosine, Tryptophan
7.	Heterocyclic amino acid	Histidine, Proline

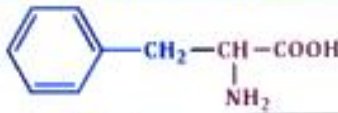
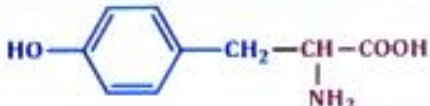
❖ **Nutritional classification of amino acid essential**

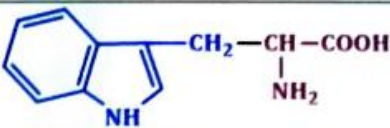
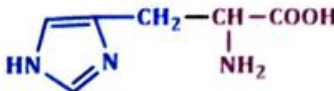
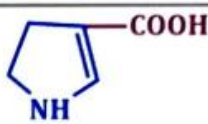
ESSENTIAL AA	NON-ESSENTIAL AA	SEMI-ESSENTIAL AA
The amino acids which cannot be synthesized by the body and need to be supplied through diet.	The amino acids which can be synthesized by human body and not essential to include in our diet.	Can be synthesized by adults but not synthesized by growing children.

Examples	Examples	Examples
V - Valine I - Isoleucine P - Phenylalanine H - Histidine A - Arginine L - Leucine L - Lysine M - Methionine T - Threonine T - Tryptophan	C - Cysteine A - Alanine T - Tyrosine A - Asparagine P - Proline A - Aspartic acid S - Serine G - Glycine G - Glutamine G - Glutamic acid	A - Arginine H - Histidine

❖ Classification of amino acids based on structure

S. NO.	AMINO ACIDS	SYMBOL	STRUCTURE
ALIPHATIC SIDE CHAIN			
1.	Glycine	Gly(G)	$\begin{array}{c} \text{H}-\text{CH}-\text{COOH} \\ \\ \text{NH}_2 \end{array}$
2.	Alanine	Ala(A)	$\begin{array}{c} \text{H}_3\text{C}-\text{CH}-\text{COOH} \\ \\ \text{NH}_2 \end{array}$
3.	Valine	Val(V)	$\begin{array}{c} \text{H}_3\text{C} \quad \text{CH}-\text{CH}-\text{COOH} \\ \quad \quad \\ \quad \text{H}_3\text{C} \quad \text{NH}_2 \end{array}$
4.	Leucine	Leu(L)	$\begin{array}{c} \text{H}_3\text{C} \quad \text{CH}-\text{CH}_2-\text{CH}-\text{COOH} \\ \quad \quad \\ \quad \text{H}_3\text{C} \quad \text{NH}_2 \end{array}$
5.	Isoleucine	Ile(I)	$\begin{array}{c} \text{C}_2\text{H}_5 \quad \text{CH}-\text{CH}-\text{COOH} \\ \quad \quad \\ \quad \text{H}_3\text{C} \quad \text{NH}_2 \end{array}$

SIDE CHAIN CONTAINING ACIDIC GROUP			
6.	Aspartic acid	Asp(D)	$\text{HOOC}-\text{CH}_2-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$
7.	Asparagine	Asn(N)	$\text{H}_2\text{N}-\overset{\text{O}}{\underset{\text{ }}{\text{C}}}-\text{CH}_2-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$
8.	Glutamic acid	Glu(E)	$\text{HOOC}-\text{CH}_2-\text{CH}_2-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$
9.	Glutamine	Gln(Q)	$\text{H}_2\text{N}-\overset{\text{O}}{\underset{\text{ }}{\text{C}}}-\text{CH}_2-\text{CH}_2-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$
SIDE CHAIN CONTAINING BASIC GROUP			
10.	Arginine	Arg(R)	$\text{NH}-\text{CH}_2\text{CH}_2-\text{CH}_2-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$ $\text{HN}=\underset{\text{NH}_2}{\text{C}}-\text{NH}_2$
11.	Lysine	Lys(K)	$\underset{\text{NH}_2}{\text{CH}_2}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$
SULPHUR CONTAINING SIDE CHAIN			
12.	Cysteine	Cys(C)	$\text{SH}-\text{CH}_2-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$
13.	Methionine	Met(M)	$\text{H}_3\text{C}-\text{S}-\text{CH}_2-\text{CH}_2-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$
SIDE CHAIN CONTAINING HYDROXYL GROUP			
14.	Serine	Ser(S)	$\text{OH}-\text{CH}_2-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$
15.	Threonine	Thr(T)	$\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$
SIDE CHAIN CONTAINING AROMATIC GROUP			
16.	Phenylalanine	Phe(F)	
17.	Tyrosine	Tyr(Y)	

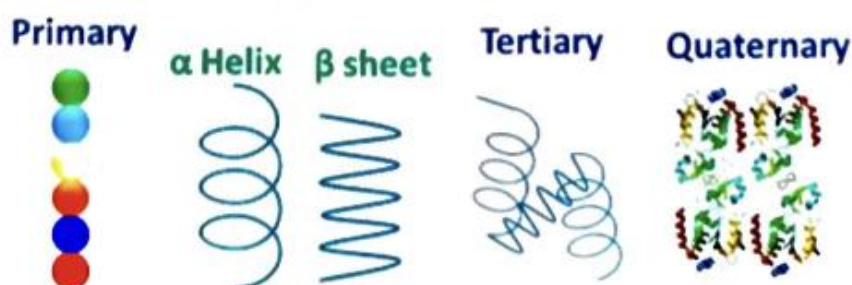
18.	Tryptophan	Trp(W)	
SIDE CHAIN CONTAINING HETEROCYCLIC GROUP			
19.	Histidine	His(H)	
20.	Proline	Pro(P)	

❖ Structure of proteins

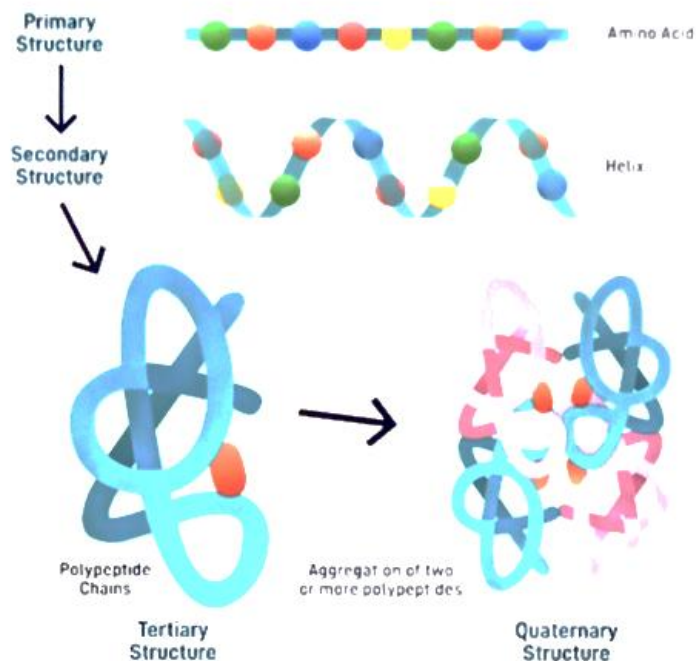
- Proteins are the **polymers of L-a-amino acids**. The structure of proteins is rather complex which can be divided into **4 levels of organization**.

- Primary structure:** The **linear sequence** of amino acids forming the backbone of proteins (**polypeptides**).
- Secondary structure:** The **spatial arrangement** of protein by twisting of the polypeptide chain.
- Tertiary structure:** The three dimensional structure of a **functional protein**.
- Quaternary structure:** Some of the proteins are composed of **two or more polypeptide chains** referred to as subunits. The **spatial arrangement** of these subunits is known as quaternary structure.

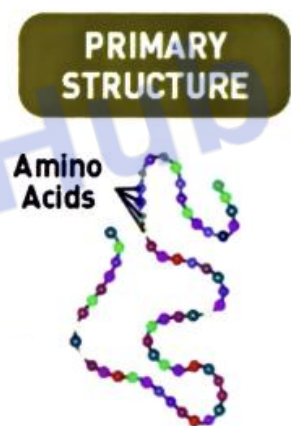
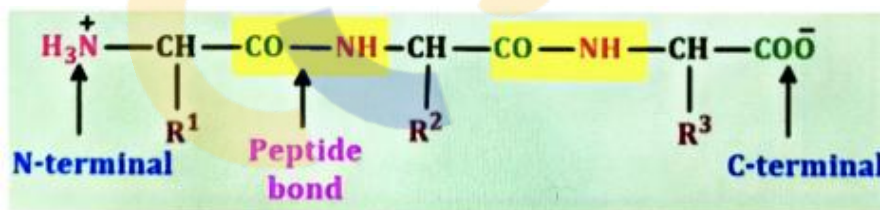
Structure of Proteins



PROTEIN STRUCTURE



- The **sequence of amino acids** forming the backbone of proteins and location of any **disulfide bond** in a protein is **called**, the primary structure of the protein.
- Simple (**linear**) amino acid sequence.
- Amino acids are joined **covalently by peptide bond**.

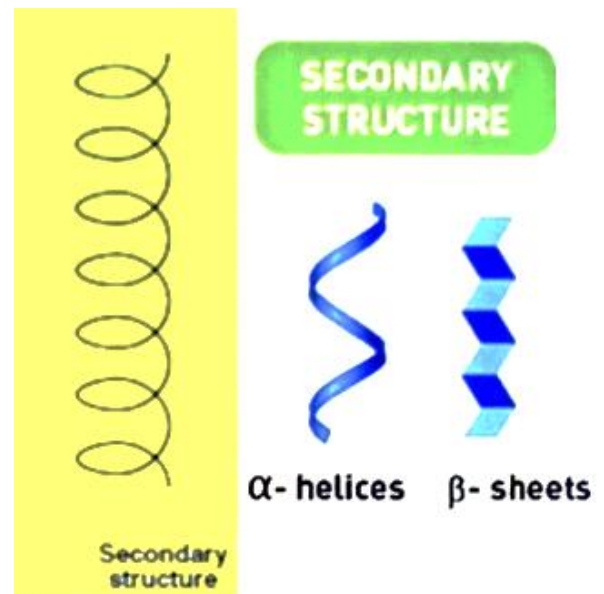


- Protein biosynthesis Start form **N-termination amino acid**.
- Determination** of amino acid sequence is by **Sanger's and Edman's reagent**.
E.g.- Insulin

❖ Secondary structure

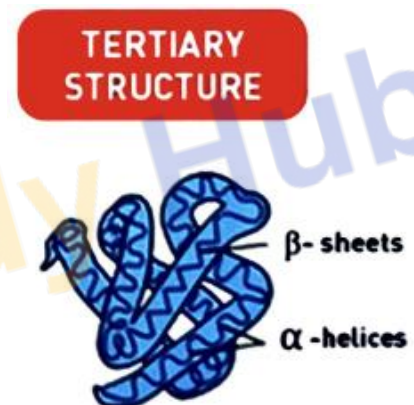
- The **spatial arrangement** of protein by **twisting of the polypeptide chain**.

- Spatial arrangement of protein by twisting of polypeptide chain.
- Consist of **α -helix** and **β pleated sheet**.
- Stabilized by **hydrogen and electrostatic bonding**.
- E.g.: **α -helical** – Leucine, Glutamate and Alanine **β -pleated sheet** – Threonine, Histidine, Tyrosine and Isoleucine



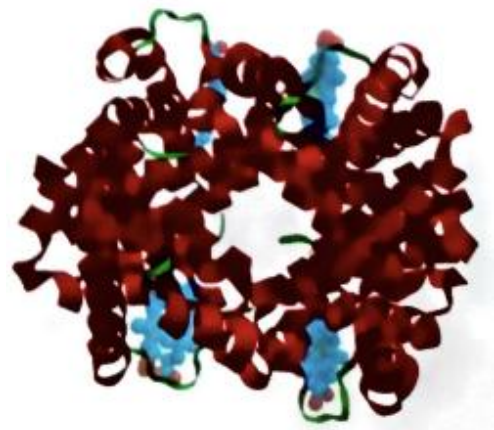
❖ Tertiary structure

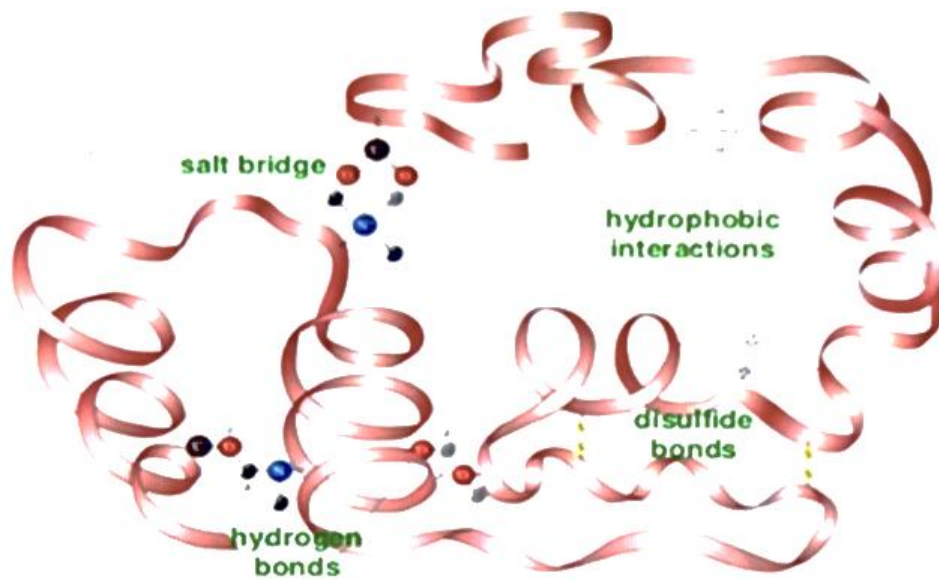
- The **three dimensional structure** of a functional protein.
- Stabilized by **hydrogen, hydrophobic bond, Van der wall force, disulfide bond, ionic bond**.
- Thermodynamically **stable**.
- E.g.: Myoglobin, Immunoglobulin



❖ Quaternary structure

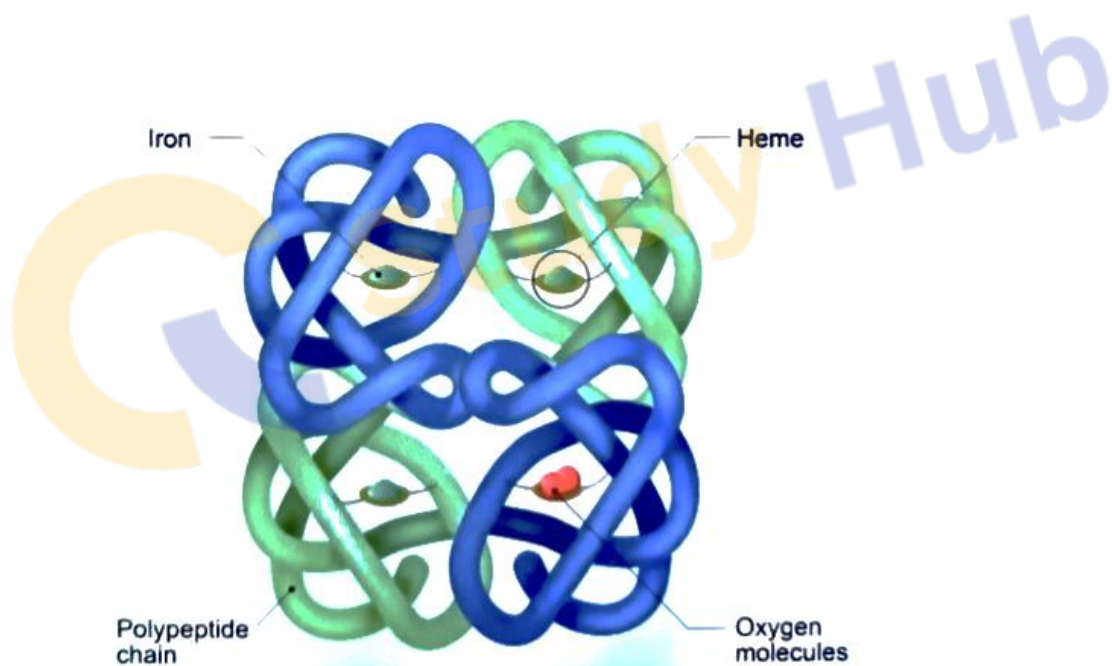
- Some of the proteins are composed of **two or more polypeptide chains** referred to as **subunits**. The **spatial arrangement** of these **subunits** is known as **quaternary structure**.





- Stabilized by hydrophobic bonds, hydrogen bonds, ionic bonds.
- E.g.- Hemoglobin, **Creatine kinase**, Lactate dehydrogenase

HEMOGLOBIN



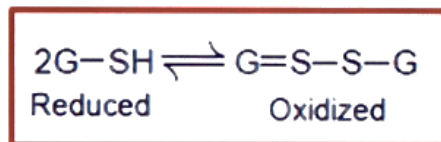
BIOLOGICALLY IMPORTANT PEPTIDES

- Several peptides occur in the living organisms that display a **wide spectrum of biological functions**.
- Generally, the term '**peptide**' is applied when the number of constituent amino acids is **less than 10**.

- Some examples of biologically active peptides and their functions are described here.

1. Glutathione :

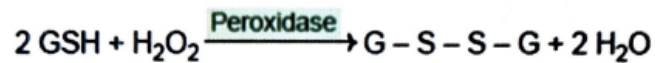
- It is a tripeptide composed of 3 amino acids. Chemically, **glutathione is γ - glutamyl-cysteinyl-glycine**. It is widely distributed in nature and exists in reduced or oxidized states.



► Functions :

- In a steady state, the cells generally maintain a ratio of about **100/1 of GSH to G-S-S-G**. The reversible oxidation reduction of glutathione is important for many of its biological functions.
- Glutathione serves as a **coenzyme for certain enzymes e.g. prostaglandin PGE₂ synthetase, glyoxalase**.
- It prevents the oxidation of **sulfhydryl (-SH) groups** of several proteins to **disulfide (-S-S-) groups**.
- This is essential for the protein function, including that of enzymes.
- It is believed that **glutathione in association with glutathione reductase participates in the formation of correct disulfide bonds** in several proteins.
- Glutathione (**reduced**) performs specialized functions in erythrocytes
 - (i) It maintains RBC membrane structure and integrity.
 - (ii) It protects hemoglobin from getting oxidized by agents such as H₂O₂.
- Glutathione is involved in the transport of amino acids in the intestine and kidney tubules **via γ -glutamyl cycle or Meister cycle**.

- **Glutathione is involved in the detoxification process.** The toxic substances (organophosphates, nitro compounds) are converted to mercapturic acids.
- Toxic amounts of peroxides and free radicals produced in the cells are scavenged by glutathione peroxidase (a selenium containing enzyme).



2. Thyrotropin releasing hormone (TRH) :

- It is a tripeptide secreted by hypothalamus.
- TRH stimulates pituitary gland to **release thyrotropic hormone**.

3. Oxytocin :

- It is a hormone secreted by **posterior pituitary gland** and contains **9 amino acids (nonapeptide)**.
- Oxytocin causes contraction of uterus.

4. Vasopressin (antidiuretic hormone, ADH) :

- ADH is also a nonapeptide produced by **posterior pituitary gland**.
- It **stimulates kidneys to retain water** and thus increases the blood pressure.

5. Angiotensin :

- Angiotensin I is a **Decapeptide (10 amino acids)** which is converted to **angiotensin II (8 amino acids)**.
- The latter has **more hypertensive effect**.
- **Angiotensin II also stimulates the release of aldosterone from adrenal gland.**

6. Methionine enkephalin :

- It is a pentapeptide found in the brain and has **opiate like function**.
- **It inhibits the sense of a pain.**

7. Bradykinin and kallidin :

- Both of them act as **powerful vasodilators**.
- They are produced from plasma proteins by **snake venom enzymes**.

8. Peptide antibiotics :

- Antibiotics such as **gramicidin, bacitracin, tyrocidine and actinomycin** are peptide in nature.

9. Aspartame :

- It is a dipeptide (aspartyl phenylalanine methyl ester), produced by a **combination of aspartic acid and phenylalanine**.

10. Gastrointestinal hormones :

- **Gastrin, secretin etc.** are gastrointestinal peptides which **serve as hormones**.
-



BIOLOGICALLY IMPORTANT COMPOUNDS FORMED BY AMINO ACID

AMINO ACID	AMINES	FUNCTIONS
Cysteine	Taurine	Constituent of bile acid
Glutamic acid	Y-Amino butyric acid (GABA)	Inhibitory neurotransmitter
Histidine	Histamine	Vasodilation, Promotes gastric HCl and pepsin synthesis
Phenylalanine, Tyrosine	Dopamine	For adrenaline, Nor adrenaline
Serine	Ethanolamine	Form Choline
Tryptophan	• Melatonin	Elevated BP stimulated, Vasoconstrictor
Tyrosine	Tyramine	Vasoconstriction (↑BP)
Methionine	Spermine	Growth, Regulator of transcription and translation
Glycine	-	Formation of Heme
Tryptophan	-	Helps in formation of vitamin like Niacin
Aspartic acid	-	Pyrimidine base
Glycine, Aspartic acid, Glutamine	-	Purine base



Neurotransmitter



Stemmed and regulation



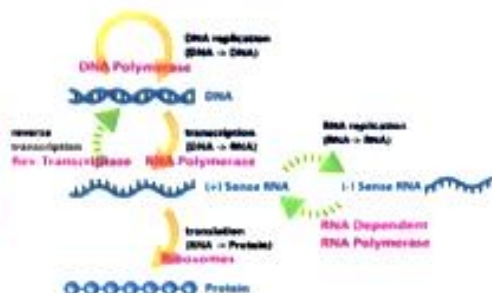
Heme



Hemoglobin



Erythrocyte



UNIT – 1: BIOENERGETICS-PART-3

Points to be covered in this topic

- ❖ Concept of free energy
- ❖ Endergonic and exergonic reaction
- ❖ Relationship between free energy, enthalpy & entropy
- ❖ Redox potential
- ❖ Energy rich compounds
- ❖ Classification
- ❖ Biological significances of ATP and Cyclic AMP

BIOENERGETICS

❖ INTRODUCTION

- The term "**Bioenergetics**" is made up of two words: Bio means Life or Living, Energetics means study of energy
- So, basically Bioenergetics is "**the study of energy changes in biological reactions**".
- Bioenergetics is the field of biochemistry concerned with the **transformation and use of energy by living cells**.

- **The goal of bioenergetics** is to describe how living organisms acquire, transform and utilize energy in order to perform biological work. The study of **metabolic pathways is thus essential to bioenergetics**.
 - The chemical reactions performed by an organism make up its metabolism.
 - **Catabolic reactions** involve the **breakdown of chemical molecules**.
 - **Anabolic reactions** involve the **synthesis of compounds**.
-

❖ Adenosine triphosphate (ATP)

- **Adenosine triphosphate (ATP)** is the main "energy currency" for organisms; the goal of metabolic and catabolic processes are:
- To **synthesize ATP** from available starting materials (from the environment), and To **break-down ATP** (into adenosine diphosphate (ADP) and inorganic phosphate) by utilizing it in biological processes.
- In a cell, the **ratio of ATP to ADP concentrations** is known as the "energy charge" of the cell. A cell can use this energy charge to relay information about cellular needs;
- If there is **more ATP than ADP available**, the cell can use ATP to do work.
- If there is **more ADP than ATP available**, the cell must **synthesize ATP via oxidative phosphorylation**.

FREE ENERGY

- Every living cell and organism must **perform work to stay alive, to grow and to reproduce**. The energy processes in living organisms are defined by the **basic laws of thermodynamics**.
- The **energy actually available to do work** (utilizable) is known as **free energy**.

- **Changes in the free energy (AG)** are valuable in predicting the **feasibility of chemical reactions**.
 - The reactions can occur spontaneously if they are accompanied by **decrease in free energy**.
 - During a chemical reaction, **heat may be released or absorbed**.
 - **Cells require a source of free energy**.
 - **Cells are isothermal systems**, meaning they function at a **constant temperature & pressure**.
 - **Photosynthetic cells acquire** free energy from absorbed solar radiation.
 - **Heterotrophic cells acquire** free energy from nutrient molecules.
 - Cells transform this **free energy into ATP & other energy-rich compounds** to provide energy for biological work.
-

❖ Principle of Bioenergetics

✓ **Biological energy transformations obey the laws of thermodynamics**

- **1st Law of Thermodynamics = Principle of conservation of energy**
 - For any physical or chemical change, the **total amount of energy in a closed system remains constant**.
 - Energy may change form or it may be transported from one region to another, but **it cannot be created or destroyed**.
 - **2nd Law of Thermodynamics = Universe tends toward increasing disorder**
 - In all natural processes, the entropy of the universe **increases (unless energy requiring processes counteract it)**.
-

❖ The important state functions for the study of biological systems are:

1. **The Gibbs free energy (G)** – equal to the **total amount of energy capable of doing work** during a process at **constant temperature and pressure**.
 - If ΔG is **negative** $\rightarrow$ process is **spontaneous** and termed **exergonic**.
 - If ΔG is **positive** $\rightarrow$ process is **non-spontaneous** and termed **endergonic**.
 - If ΔG is **equal to zero** $\rightarrow$ process has reached **equilibrium**.
2. **The Enthalpy (H)** – heat content of the system. Enthalpy is the amount of heat energy transferred (heat absorbed or emitted) in a chemical process under constant pressure.
 - When ΔH is **negative** $\rightarrow$ process **produces heat** and is termed **exothermic**.
 - When ΔH is **positive** $\rightarrow$ process **absorbs heat** and is termed **endothermic**.
3. **The Entropy (S)** – a **quantitative expression of the degree of randomness or disorder** of the system. Entropy measures the amount of heat dispersed or transferred during a chemical process.
 - When ΔS is **positive** $\rightarrow$ disorder of the system has **increased**.
 - When ΔS is **negative** $\rightarrow$ disorder of the system has **decreased**.

RELATIONSHIP BETWEEN THE CHANGE IN FREE ENERGY, ENTHALPY AND ENTROPY

- The conditions of biological systems are **constant temperature and pressure**.

- Under such conditions, the **relationships between the change in free energy, enthalpy and entropy** can be described by the expression where **T** is the **temperature of the system in Kelvin**.

$$\Delta G = \Delta H - T\Delta S$$

Where:

- ΔG** = Gibbs Free Energy
- ΔH** = Change in Enthalpy
- T** = Temperature in K
- ΔS** = Change in Entropy
- T** represents the absolute temperature in **Kelvin** ($K = 273 + ^\circ C$).

REDOX POTENTIAL

- The **oxidation-reduction potential** may be defined as a **quantitative expression of the tendency** that a compound has to give or **receive electrons**.
- The **redox potential** of a system may be calculated from the following equation:

$$E = E_0 + \frac{0.0592}{n} \log \frac{\text{Conc. of Reducing agent}}{\text{Conc. of Oxidising agent}}$$

- In Bioenergetics **Redox Potential** is the ratio of NAD^+ to $\text{NADH} + \text{H}^+$.
- It describes the **availability of NAD^+** for metabolism.

◆ Types of Bioenergetics Reactions

EXERGONIC REACTION

- Exergonic implies the **release of energy** from a **spontaneous chemical reaction** without any concomitant utilization of energy.

- These reactions have an ability to perform work and **include most of the catabolic reactions in cellular respiration.**
- Most of these reactions involve the **breaking of bonds during the formation of reaction intermediates.**
- The release of free energy, G , in an **exergonic reaction** (at constant pressure and temperature) is denoted as:

$$\Delta G = G_{\text{products}} - G_{\text{reactants}} < 0 \quad [\text{i.e. } \Delta G = \text{negative}]$$

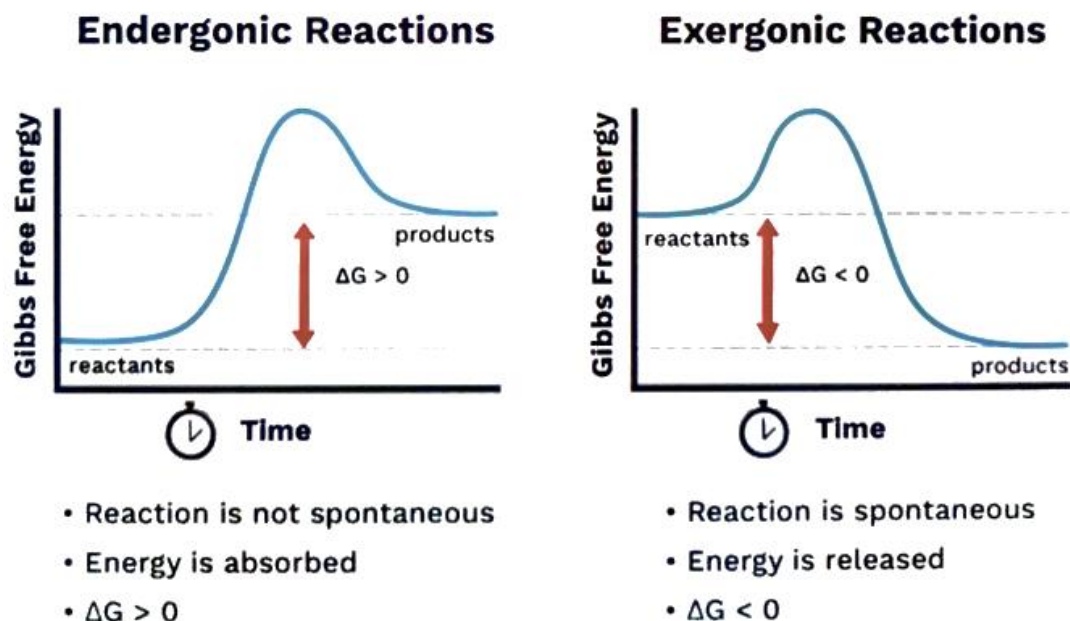
- In **exergonic reactions**, energy is released to the surrounding. Due to that reason, the **change in enthalpy is a negative value for exergonic reactions.**
- The entropy is **increased due to the disorder of the system.**
- Exergonic reactions include exothermic reactions.

ENDERGONIC REACTIONS

- **Endergonic** in turn is the opposite of **exergonic** in being non-spontaneous and requires an input of free energy.
- Most of the anabolic reactions like **photosynthesis** and **DNA** and **protein synthesis** are **endergonic** in nature.
- The release of free energy, G , in an **endergonic reaction (at constant pressure & temperature)** is denoted as

$$\Delta G = G_{\text{products}} - G_{\text{reactants}} > 0 \quad [\text{i.e. } \Delta G = \text{positive}]$$

- In a **non-spontaneous reaction**, energy should be provided from outside for the progression of the reaction.
- Since new products are formed, the **entropy** of the system is decreased.
- Then, according to the above equation, the **ΔG** is a positive value.
- **Endergonic reactions** include **endothermic reactions.**



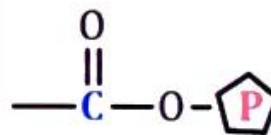
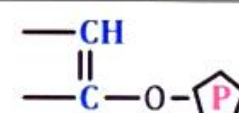
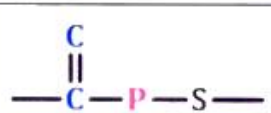
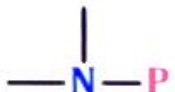
❖ Difference between Endergonic and Exergonic Reactions

Endergonic	Exergonic
Endergonic is a type of reaction that has a positive Gibbs free energy	Exergonic is a type of reaction that has a negative Gibbs free energy
Gibbs free energy is a positive value	Gibbs free energy is a negative value
The energy of reactants is lower than that of the products	The energy of reactants is higher than that of the products
Entropy is decreased	Entropy is increased
Reactions are non-spontaneous	Reactions are spontaneous
Endothermic reactions are endergonic	Exothermic reactions are exergonic
Always require energy to begin the reaction	Absorb energy from the surrounding
Do not need energy to begin the reaction	Release energy to the surrounding

HIGH ENERGY COMPOUNDS

- Certain compounds are encountered in the biological system which, on **hydrolysis**, yields energy.
- The term **high-energy compounds** or **energy rich compounds** are usually applied to substances which possess sufficient free energy to liberate at least **7 Cal/mol** at **pH 7.0**.
- Certain other compounds which liberate **less than 7.0 Cal/mol** (lower than **ATP hydrolysis to ADP + Pi**) are referred to as **low-energy compounds**.
- All the **high-energy compounds** when hydrolyzed liberate more energy than that of **ATP**.
- Most of **high energy compounds** contain **phosphate group** (exception **acetyl CoA**) hence they are also called **high energy phosphates**.

CLASSIFICATION OF HIGH ENERGY COMPOUNDS

High Energy Compounds		
Class	Bond	Example (s)
Pyrophosphates		ATP, pyrophosphate
Acyl phosphates		1,3-Bisphosphoglycerate, Carbamoyl phosphate, Acetyl phosphate.
Enol phosphates		Phosphoenol pyruvate
Thiol esters (thioesters)		Acetyl CoA, Acyl CoA
Guanido phosphates (phosphagen)		Phosphocreatine, Phosphoarginine

ADENOSINE TRIPHOSPHATE (ATP)

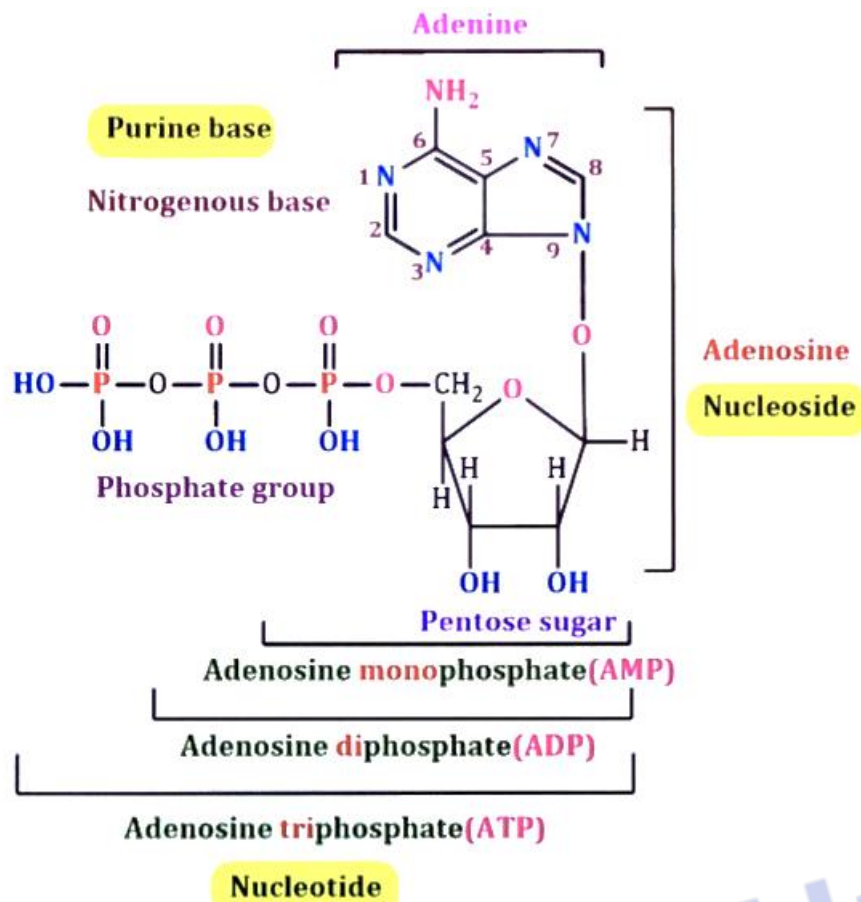
- **Adenosine-5'-triphosphate (ATP)** is a multifunctional nucleotide **used in cells as a coenzyme**.
- It is often called the "**molecular unit of currency**" of **intracellular energy transfer**. ATP transports chemical energy within cells for metabolism.
- It is produced by **photophosphorylation** and **cellular respiration** and used by enzymes and structural proteins in many cellular processes.
- One molecule of **ATP contains three phosphate groups** and it is produced by **ATP synthase** from **Inorganic Phosphate** and **Adenosine Diphosphate (ADP)** or **Adenosine Monophosphate (AMP)**.

The three main functions of ATP in cellular function:

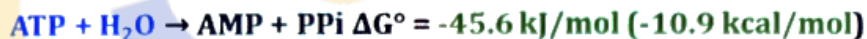
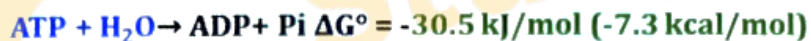
- Transporting organic substances-such **sodium, calcium, potassium-** through the cell membrane.
- Synthesizing chemical compounds, such as **protein** and **cholesterol**.
- Supplying energy for mechanical work, such as **muscle contraction**.

❖ STRUCTURE OF ATP

- The **structure** of this molecule consists of a **purine base (adenine)** attached to the **1' carbon atom** of a pentose sugar (**ribose**).
- Three **phosphate groups** are attached at the **5' C atom** of the pentose sugar.
- It is the addition and removal of these **phosphate groups** that inter-convert **ATP, ADP, and AMP**.



- The energy released by cleaving either a **phosphate (Pi)** or **pyrophosphate (PPi)** unit from ATP at standard state of 1 M are:



- These values can be used to calculate the change in energy under physiological conditions and the cellular **ATP/ADP ratio** (also known as the **Energy Charge**).

❖ Biological significance of ATP

- ATP is the most important **energy giving compound** found within all living cells.
- It is used in **impulse formation** and supplies energy needed for **cell metabolism**.

- It adds energy in the **contraction of muscles**.
 - It serves as a processes **source of energy** for **Catabolic** and **Anabolic processes**.
 - **ATP** captures chemical energy obtained from the **breakdown of food molecules** and releases it as fuel for **cellular processes**.
 - **ATP** receives energy from **exergonic reactions** and **transfers this energy to endergonic reactions**.
 - During **photosynthesis** and **respiratory cycle**.
 - It **catalyzes the reactions** by **providing required energy** to produce essential substances (**nutrients**).
-

❖ **Biological significance of cAMP**

- **cAMP** works as a **secondary messenger**, which is used for **intracellular signal transduction** which involves the transferring into cells the effects of hormones like **adrenaline** and **glucagon**. The hormones are unable to pass through **plasma membrane**. This is the reason why they require **secondary messenger** like **cAMP**.
- **cAMP** binds to **ion channels** like **HCN** and regulates the passage of specific ion through it across the membrane.
- **cAMP** plays a role in the **activation of the protein kinase enzymes** which participate in biochemical regulation of **glycogen, carbohydrate and lipid metabolism** (These reactions are called as **pKA** or **protein kinase A dependent functions of cAMP**).
- **Activation of a Ca-channel**, which involves **activation of growth hormone. Releasing hormone** to release the **growth hormone** occurs through the action of **cAMP**. However, this function of **cAMP** does not involve any mediation or participation of **pKA** or **protein kinase A**.